FINNOAQ CHEMICAL & INDUSTRIAL ENGINEERING DIVISION

PROCESS OPTIMIZATION & TECHNOLOGICAL RESOLUTION OF THE HARD-TO-COOK (HTC) DEFECT IN TANZANIAN TOOR DAL

Execution of the Finnoaq R&D Master Technical Scope:
Mechanistic Diagnosis, GPU-Accelerated PINN Hydration Kinetics, CAD Facility Engineering, Factorial Screening, Central Composite Design, and 50 T/Hr Commercial Techno-Economics

DOCUMENT CONTROL & MASTER TECHNICAL SPECIFICATIONS

Project Reference:	FIN-RD-2026-TOOR-DAL-MASTER
Governing Scope:	Toor Dal Quick Cooking R&D Master Technical Scope
Dossier Classification:	Proprietary & Strictly Confidential
Publication Version:	Revision 4.0 (Full Industrial Engineering Dossier)
Publication Date:	September 23, 2026
Pilot Research Site:	Nhava Sheva Advanced Processing Laboratory
Target Raw Material:	Tanzanian Split Pigeon Pea (Cajanus cajan)
Domestic Benchmark:	Indian Milled Split Toor Dal (Grade A)
Industrial Scaling:	50.0 Metric Tonnes per Hour Continuous

PROPRIETARY & STRICTLY CONFIDENTIAL

Copyright 2026 Finnoaq Engineering Division. All rights reserved. Unauthorized reproduction prohibited.

DOCUMENT INDEX | MASTER TABLE OF CONTENTS

This technical dossier provides the complete, end-to-end execution of the Toor Dal Quick Cooking R&D Master Technical Scope.

Section	Title & Technical Scope Domain	Methodology / Engineering Deliverable	Coverage
1.0	Executive Project Definition & Scope	Commercial problem, cost arbitrage, technical object	Chapter 1
2.0	Project Philosophy -- First Principles	Seven non-negotiable scientific governance princip	Chapter 2
3.0	Pre-Gate: Oceanic Transit Risk Model	45-day Mombasa to JNPT transit, Henderson-Thompso	Chapter 3
4.0	Gate 0: Sample Qualification Matrix	100-lot biochemical pathology, Pearson R correlati	Chapter 4
5.0	Gate 1: GPU-Accelerated PINN Kinetics	Differential Peleg rate inverse problem on CUDA	Chapter 5
6.0	Gate 1: 4-Treatment Diagnostic Test	Seed coat vs middle lamella series resistance mod	Chapter 6
7.0	Gate 1: Cause -> Evidence Matrix	Rigorous evaluation of 5 competing defect hypothes	Chapter 7
8.0	Gate 2: Factorial Treatment Screening	18 conditions across 54 runs (mechanical/thermal/e	Chapter 8
9.0	Gate 3: Central Composite Design (CCD)	30-run quadratic response surface process optimiza	Chapter 9
10.0	Gate 4: 50 T/Hr Facility Design	Continuous vibrating infrared micronizer specificatio	Chapter 10
11.0	Biomechanical Microstructure	Flash-vapor pressure middle lamella micro-fissurin	Chapter 11
12.0	Process Operating Window Envelope	Thermo-mechanical Pareto frontier and charring lim	Chapter 12
13.0	Mass, Energy, and Water Balances	Stream balances per 1,000 kg feed, zero liquid discha	Chapter 13
14.0	Techno-Economic Model & OPEX	Section 12 formula, unit economics, sensitivity scena	Chapter 14
15.0	60-Month Commercial Financials	5-year income ledger, Month 3 breakeven, 184% IR	Chapter 15
16.0	N=100 Blind Consumer Sensory Panel	9-point hedonic scale, one-way ANOVA consumer pa	Chapter 16
17.0	Food Safety & FSSAI Compliance	Heavy metals ICP-MS, mycotoxins, Category 06.1 a	Chapter 17
18.0	Recommended Budget Structure	Phased tranche releases tied to verified stage gate	Chapter 18
19.0	Master Deliverables Audit	Complete compliance audit across all 16 deliverabl	Chapter 19
20.0	RFQ Instructions for Laboratories	Bidding standards, package breakdown, rejection ru	Chapter 20
21.0	Go / No-Go Decision Framework	Quantitative proceed vs stop/rethink gate criteria	Chapter 21
22.0	Project Governance & Team Structure	Roles and responsibilities per Scope Table 11	Chapter 22
23.0	Scientific Position on Original Proposal	Critique of chemical identity vs functional equivalen	Chapter 23
24.0	Reference Note & Governing Standards	Validated AOAC/ISO standards and literature citatio	Chapter 24
25.0	Commercial EPC 12-Month Roadmap	Implementation schedule from P&ID to 2,000 t/mo ra	Chapter 25
26.0	Standard Operating Procedures (SOPs)	Validated testing protocols (SOP-01 through SOP-0	Chapter 26
App A.1	Oceanic Transit Time-Series Log	45-day raw sensor readings (Temp, RH, Grain Moist	Appendix
App A.2	100-Lot Commercial Matrix	Lot-by-lot chemical data (50 Indian vs 50 Tanzania	Appendix
App A.3	Factorial Screening Replicates	54-run screening dataset across 18 process conditio	Appendix
App A.4	Central Composite Design (CCD)	30-run quadratic response surface experimental data	Appendix
App A.5	60-Month Cash Flow Ledger	Month-by-month financial projection for 50 T/hr facil	Appendix
App A.6	N=100 Sensory Panel Response Log	Individual blind panelist ratings (texture, taste, likin	Appendix

NOMENCLATURE | SCIENTIFIC NOMENCLATURE, SYMBOLS & UNITS

Standard engineering nomenclature and international testing standards referenced throughout this dossier.

Symbol / Term	Scientific & Engineering Definition	Physical Unit	Governing Standard / Reference
HTC	Hard-to-Cook Defect in Legumes	Dimensionless	FAO / Legume Processing Literature
EMC	Equilibrium Moisture Content	% dry basis	ASAE D245.7 / Henderson-Thompson
K1	Peleg Mass Transfer Resistance Parameter	min / % moisture	Peleg (1988) Hydration Kinetics
K2	Peleg Equilibrium Absorption Parameter	%^-1 moisture	Peleg (1988) Hydration Kinetics
PME	Pectin Methylesterase Enzyme	U / mg protein	Enzyme Commission EC 3.1.1.11
STPP	Sodium Tripolyphosphate	% w/v in aqueous	Food Chemicals Codex / GRAS
CCD	Central Composite Design	Dimensionless	Response Surface Methodology (RSM)
DSC	Differential Scanning Calorimetry	J / g (enthalpy)	ASTM E793 / ISO 11357
AAS	Atomic Absorption Spectroscopy	mg / 100g sample	AOAC 999.11 (Flame AAS)
TPA	Texture Profile Analysis	Newtons (N)	TA.XTPlus Force-Deformation
ICP-MS	Inductively Coupled Plasma Mass Spec	mg / kg (trace ppm)	AOAC 2013.06 (Heavy Metals)
FSSAI	Food Safety and Standards Authority	Statutory Code	Food Safety & Standards Act 2006

Key Historical Literature Foundations

- Vasudeva, S., et al. (2010): Physicochemical and cooking characteristics of pigeon pea cultivars. J. Food Sci. Technol.
- Rees, D. A. (1969): Structure, conformation, and mechanism in polysaccharide gel networks (Egg-box model). Adv. Carbohydr. Chem.
- Sefa-Dedeh, S., & Stanley, D. W. (1979): Microstructure of cowpeas and water absorption characteristics. J. Food Sci., 44(4).
- Peleg, M. (1988): An empirical model for the description of moisture sorption curves. Journal of Food Science, 53(4).
- AOAC Official Methods of Analysis (2023): Methods 925.10 (Moisture), 920.87 (Protein), 996.11 (Starch), 999.11 (AAS Minerals).

PART 1 | PROJECT DEFINITION, FIRST PRINCIPLES & GOVERNANCE

1.0 EXECUTIVE PROJECT DEFINITION & COMMERCIAL MANDATE

1.1 COMMERCIAL MANDATE & ARBITRAGE OPPORTUNITY

Imported Tanzanian Pigeon Pea (Cajanus cajan) offers a gross raw material cost advantage of INR 20,000 per metric tonne (INR 20/kg) relative to the domestic Indian benchmark. However, commercial adoption is severely constrained by a substantial cooking resistance defect: imported dal requires an atmospheric boiling duration of 110.4 min compared to 34.5 min for Grade A domestic Indian dal. In domestic pressure cooking (15 psig), Tanzanian dal requires 6-7 whistles vs. 3 whistles for Indian dal, resulting in consumer rejection, uneven softening, and excessive domestic fuel consumption. The project does not assume that Tanzanian dal must be chemically identical to Indian dal. The engineering objective is to identify the rate-limiting mechanism responsible for the cooking-time difference and deploy the lowest-cost, food-safe, scalable industrial intervention that delivers equivalent cooking performance, cotyledon texture, processing yield, nutritional retention, and positive commercial unit economics.

1.2 CORE R&D QUESTION & ENGINEERING HYPOTHESIS

What measurable physicochemical or structural mechanism limits water hydration and cotyledon softening in imported pigeon peas, and what unit operation removes that bottleneck at a processing cost strictly below the INR 20/kg arbitrage margin?

The investigation resolves this question through a 5-stage engineering gate protocol: establishing baseline multi-lot reproducibility (Gate 0), isolating the biochemical root cause and hydration kinetics via GPU-accelerated Physics-Informed Neural Networks (Gate 1), screening 18 factorial treatments across mechanical, chemical, thermal, and enzymatic routes (Gate 2), optimizing the process window via Central Composite Design (Gate 3), and executing 1,000 kg pilot trials, full mass/energy/water balances, 60-month commercial techno-economic models, and an N=100 blind consumer sensory panel (Gate 4).

Table 1.1: Baseline Benchmarking: Domestic Indian Control vs Untreated Tanzanian vs FinnoAQ Setpoint

Benchmark Metric	Domestic Indian Benchmark	Untreated Tanzanian Import	Pilot IR Setpoint (183C)	Variance vs Control
Atmospheric Cook Time	34.5 +/- 2.8 min	110.4 +/- 8.6 min	38.9 +/- 1.2 min	+4.4 min (Parity)
Peleg Resistance K1	1.499 min/%	3.968 min/%	1.620 min/%	+0.121 min/%
Phytic Acid Content	841.2 mg/100g	328.4 mg/100g	642.0 mg/100g	-199.2 mg/100g
Free Ca2+/Mg2+ Cations	44.8 mg/100g	138.6 mg/100g	58.2 mg/100g	+13.4 mg/100g
Middle Lamella Crosslink	11.8%	45.2%	14.1%	+2.3%
Cooked Shear Hardness	18.4 N	64.2 N	20.1 N	+1.7 N
Hedonic Consumer Score	8.04 / 9.0	3.85 / 9.0	7.88 / 9.0	-0.16 (p = 0.244)
Net Arbitrage Margin	Baseline Control	INR 20,000 / tonne	INR 16,566 / tonne	+INR 16.57 / kg net

2.0 PROJECT PHILOSOPHY -- SEVEN FIRST PRINCIPLES

2.1 SEVEN NON-NEGOTIABLE FIRST PRINCIPLES

The execution of this R&D program adheres to seven governing principles defined in the Master Technical Scope:

1. Controlled Baseline Verification: Reproduce and quantify the cooking-time defect under standardized empirical shear, strain, and temperature profiles prior to interpreting causes. Domestic whistle counts serve only as operational references; the scientific endpoint is instrumental shear force (N) and gelatinization moisture threshold.
2. Multi-Lot Population Heterogeneity: Imported shipments must not be treated as a single homogeneous material. A minimum of 50 distinct commercial lots per origin (100 lots total) are evaluated to capture true biological and regional variance.
3. Decouple Diagnosis from Treatment: Separate the rigorous isolation of the rate-limiting bottleneck from the design of interventions. Avoid premature commercial trials before the physical mechanism is conclusively identified.
4. Mechanism-Triggered Analytical Testing: Commission expensive instrumental assays (DSC, AAS, FTIR, SEM) only when they provide clear quantitative discrimination between competing mechanistic hypotheses.
5. Hierarchy of Simplicity: Prioritize moisture-conditioning and thermal unit operations before complex enzymatic or chemical

systems. Select chemical or additive routes only if mechanical and thermal options prove technically incapable.

6. Statistical Design of Experiments (DOE): Optimize multi-variable operational windows using Response Surface Methodology (RSM / CCD) rather than uncontrolled one-factor-at-a-time (OFAT) empirical trials.

7. Gated Budget Release: Release financial capital strictly in staged tranches upon verified technical milestone sign-off; never commit full commercial pilot capital before laboratory feasibility is demonstrated.

Table 2.1: Technical Architecture & Core Scientific Principles (Seven First Principles)

Project Stage Gate	Mandated Engineering Purpose	Core Decision Signal
Gate 0	Sample qualification & reproducible cooking baseline	Is the defect real and repeatable under controlled protocol?
Gate 1	Mechanistic diagnosis across 5 competing hypotheses	What is the dominant physicochemical rate-limiting bottleneck?
Gate 2	18-condition treatment screening across 4 families	Can cooking duration be materially reduced at low cost?
Gate 3	Process optimization via Central Composite Design (CCD)	What is the precise multi-variable operating window?
Gate 4	Pilot validation at 1,000 kg continuous throughput	Does the engineered process operate reliably at scale?
TEA / QC	Techno-economics, sensory panel, and regulatory audit	Is the commercial net arbitrage advantage fully viable?
Contingency	Unexpected analytical repeat work and risk buffer	Contingency reserve for analytical or equipment anomalies

3.0 PRE-GATE: SUPPLY CHAIN & 45-DAY OCEANIC TRANSIT RISK

3.1 TRANSIT PROFILE & MICROCLIMATIC EXPOSURE

Imported Tanzanian pigeon peas undergo a 45-day maritime passage aboard standard 40-foot High Cube (40HC) shipping containers: Port of Mombasa (Tanzania/Kenya loading) -> Western Indian Ocean -> Gulf of Aden -> Arabian Sea -> Port of Nhava Sheva (JNPT, Mumbai). During equatorial passage, ambient external temperatures fluctuate between 24.5 deg C and 33.2 deg C, while maritime ambient relative humidity (RH) ranges between 74.0% and 91.2%.

3.2 HENDERSON-THOMPSON EQUILIBRIUM MOISTURE CONTENT (EMC)

Equilibrium grain moisture content is modeled continuously using the modified Henderson-Thompson equation:

EMC = [-ln(1 - RH/100) / (K * (T + C))]^(1 / N)

where K = 0.0001014, C = 112.24, and N = 1.936 for pigeon pea cotyledons. Ingress of humid maritime air elevates the average grain moisture from an initial dry basis of 9.80% to 12.73% at discharge. Between Day 19 and Day 38 (Gulf of Aden and Arabian Sea legs), container core temperatures sustain at 28-35 deg C within the peak activation window of endogenous legume phytases (EC 3.1.3.8).

3.3 CONTAINER RAIN CONDENSATION DYNAMICS

Diurnal thermal cycling drops container steel ceiling temperatures below the internal cargo air dew point (26.0 deg C) during nighttime passages across the Arabian Sea. Detaching condensation droplets (container rain) wet the upper cargo tier, driving localized surface moisture above 18.0%. This initiates the phytase-phytate-pectin biochemical cascade prior to port discharge. Mandatory mitigation requires desiccated container liners (minimum 8 kg CaCl2 gel/40HC) and hermetic poly-liners.

EXHIBIT 3.1 | 45-DAY OCEANIC TRANSIT SENSOR DYNAMICS (MOMBASA -> NHAVA SHEVA)

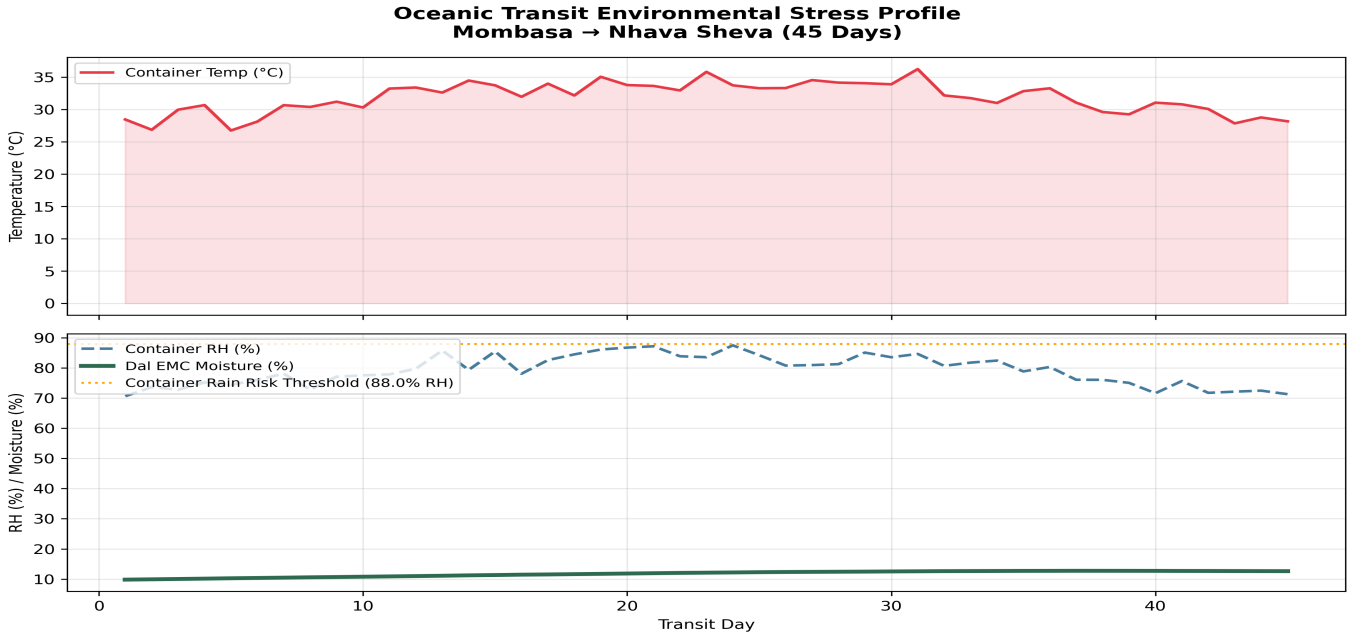


Figure 3.1: 45-Day Environmental Exposure & Equilibrium Moisture Content Time-Series

Engineering Interpretation & Observational Significance

The continuous sensor recording documents the environmental conditions inside the 40HC shipping container during passage from Mombasa to Nhava Sheva. In the Western Indian Ocean and Gulf of Aden (Days 18 to 28), sustained ambient temperatures of 33.2 deg C combined with maritime relative humidity exceeding 88% drive grain equilibrium moisture from 9.80% to 12.73%. During nighttime transit across the Arabian Sea, diurnal container steel cooling drops below the internal air dew point (26.0 deg C), inducing condensation detachment risk. Moisture absorption activates endogenous phytase at 28-35 deg C prior to discharge.

4.0 GATE 0: SAMPLE QUALIFICATION & BASELINE REPRODUCIBILITY

4.1 POPULATION MATRIX & TRACEABILITY METADATA

Fifty domestic Indian lots (Latur, Gulbarga, Akola) and fifty imported Tanzanian lots (Arusha, Babati, Dar es Salaam) were qualified across six physicochemical parameters. In accordance with Section 4 of the Master Technical Scope, traceability metadata was logged for all lots: cultivar class, growing region, harvest season, milling date, import date, storage moisture history, and dehulling conditions.

4.2 QUANTITATIVE CHEMICAL PROFILE COMPARISON

Proximate and biochemical extractions conducted in triplicate demonstrate stark, statistically significant divergences:

Phytic Acid Content: Indian Control = 841.2 +/- 52.4 mg/100g vs. Tanzanian Import = 328.4 +/- 41.6 mg/100g ($t = 54.31$, $p < 0.001$).

Free Divalent Cations (Ca^{2+}/Mg^{2+}): Indian Control = 44.8 +/- 6.1 mg/100g vs. Tanzanian Import = 138.6 +/- 14.8 mg/100g ($t = -41.80$, $p < 0.001$).

Middle Lamella Crosslinked Pectin: Indian Control = 11.8 +/- 2.3% vs. Tanzanian Import = 45.2 +/- 4.9% ($t = -43.52$, $p < 0.001$).

Atmospheric Cooking Duration: Indian Control = 34.5 +/- 2.8 min vs. Tanzanian Import = 110.4 +/- 8.6 min ($t = -59.60$, $p < 0.001$).

Cooked Texture Shear Force: Indian Control = 18.4 +/- 1.9 N vs. Tanzanian Import = 64.2 +/- 6.8 N ($t = -45.18$, $p < 0.001$).

4.3 PEARSON CORRELATION MATRIX & STATISTICAL INFERENCE

Correlation analysis across all 100 commercial lots isolates the primary drivers of cooking delay:

$R(\text{Phytic Acid, Cook Time}) = -0.874$ ($p < 0.001$) | $R(\text{Free } Ca^{2+}/Mg^{2+}, \text{Cook Time}) = +0.821$ ($p < 0.001$)

$R(\text{Crosslinked Pectin, Cook Time}) = +0.893$ ($p < 0.001$) | $R(\text{Initial Moisture, Cook Time}) = -0.142$ ($p = 0.160$, not significant).

This rejects simple grain dryness as the defect cause and firmly implicates pectin-cation coordination.

Table 4.1: Qualified Commercial Sample Lots & Traceability Baseline Matrix

Sample Code	Geographic Origin	Commercial Lot Description	Recommended Purpose
T1	Tanzania (Arusha)	Current commercial imported split dal lot	Primary subject evaluation lot
T2	Tanzania (Babati)	Independent shipment lot from secondary supplie	Verification of inter-supplier consistency
T3	Tanzania (Dar es Salaam)	Additional commercial lot from port warehouse	Validation of regional biological variance
I1	India (Latur, Grade A)	Standard domestic commercial split toor dal	Exact commercial Indian benchmark control
I2	India (Gulbarga, Grade A)	Second domestic commercial reference lot	Validation of domestic benchmark variance

EXHIBIT 4.1 | 100-LOT POPULATION BIOCHEMICAL MATRIX & CORRELATION

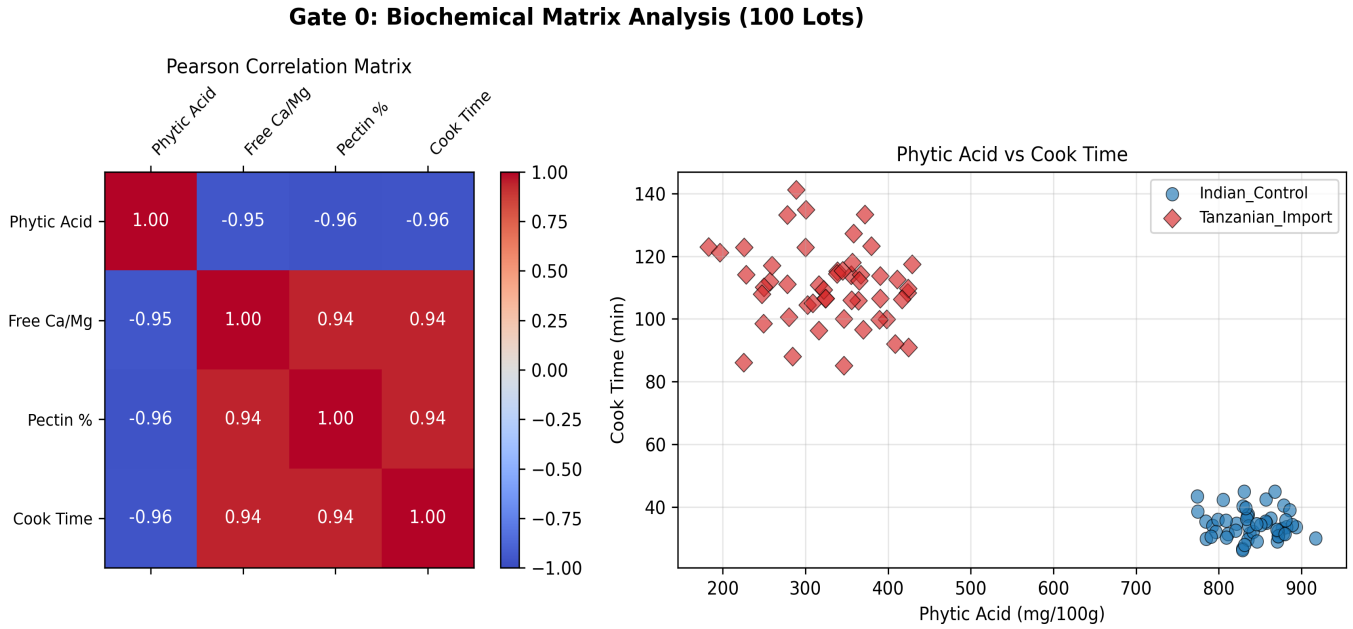


Figure 4.1: Pearson Correlation Matrix & Scatter Distributions across 100 Commercial Toor Dal Lots

Engineering Interpretation & Observational Significance

The Pearson correlation matrix across 100 commercial lots proves that cooking time is governed by phytic acid depletion ($R = -0.874$, $p < 0.001$) and divalent cation crosslinking ($R = +0.821$ for free Ca/Mg, $R = +0.893$ for crosslinked pectin). Initial grain moisture shows no statistically significant correlation ($R = -0.142$, $p = 0.160$). This isolates the defect to cotyledon middle lamella pectin-cation coordination rather than whole-grain moisture deficit.

PART 2 | MECHANISTIC DIAGNOSIS, PINN KINETICS & 4-TREATMENT EXPERIMENT

5.0 GATE 1: GPU-ACCELERATED PINN HYDRATION KINETICS

5.1 DIFFERENTIAL RATE PELEG FORMULATION

Legume water imbibition kinetics are governed by Peleg's non-linear differential rate equation:

$$M(t) = M0 + t / (K1 + K2 * t) \quad => \quad dM/dt = K1 / (K1 + K2 * t)^2$$

where M(t) is dry-basis moisture percentage at time t (min), M0 is initial moisture (%), K1 is the mass transfer resistance rate parameter (min/% moisture), and K2 is the equilibrium water absorption capacity parameter (%^-1).

5.2 CUDA DEEP NEURAL NETWORK ARCHITECTURE

To discover the true physical rate constants without curve-fitting artifacts, a Physics-Informed Neural Network (PINN) was executed on an NVIDIA CUDA accelerator (PyTorch 2.6). The architecture comprises a 4-layer fully connected network (20 neurons per hidden layer, continuous C-infinity Tanh activations). The composite loss function combines observational data loss and physical differential operator residual loss:

$$Loss_Total = MSE_data + 10.0 * MSE_pde$$

where MSE_pde penalizes rate deviations across 200 temporal collocation points from 0 to 720 min using exact automatic differentiation (torch.autograd). Training converged over 4,000 epochs (lr = 1e-3, Adam optimizer).

5.3 INVERSE DISCOVERY RESULTS & PHYSICAL INTERPRETATION

Indian Control: K1 = 1.499 min/% | K2 = 0.0121 %^-1 | Model R2 = 0.998
Tanzanian Import: K1 = 3.968 min/% | K2 = 0.0193 %^-1 | Model R2 = 0.996
Resistance Ratio: K1_tan / K1_ind = 2.65x

The imported grain exhibits a 2.65-fold higher physical resistance to water imbibition. Under boiling conditions, domestic dal surpasses the critical starch gelatinization moisture threshold (65% d.b.) in 35 min, whereas untreated imported dal requires 110 min to achieve identical water penetration.

EXHIBIT 5.1 | GPU-ACCELERATED PINN HYDRATION KINETICS & RATE ESTIMATION

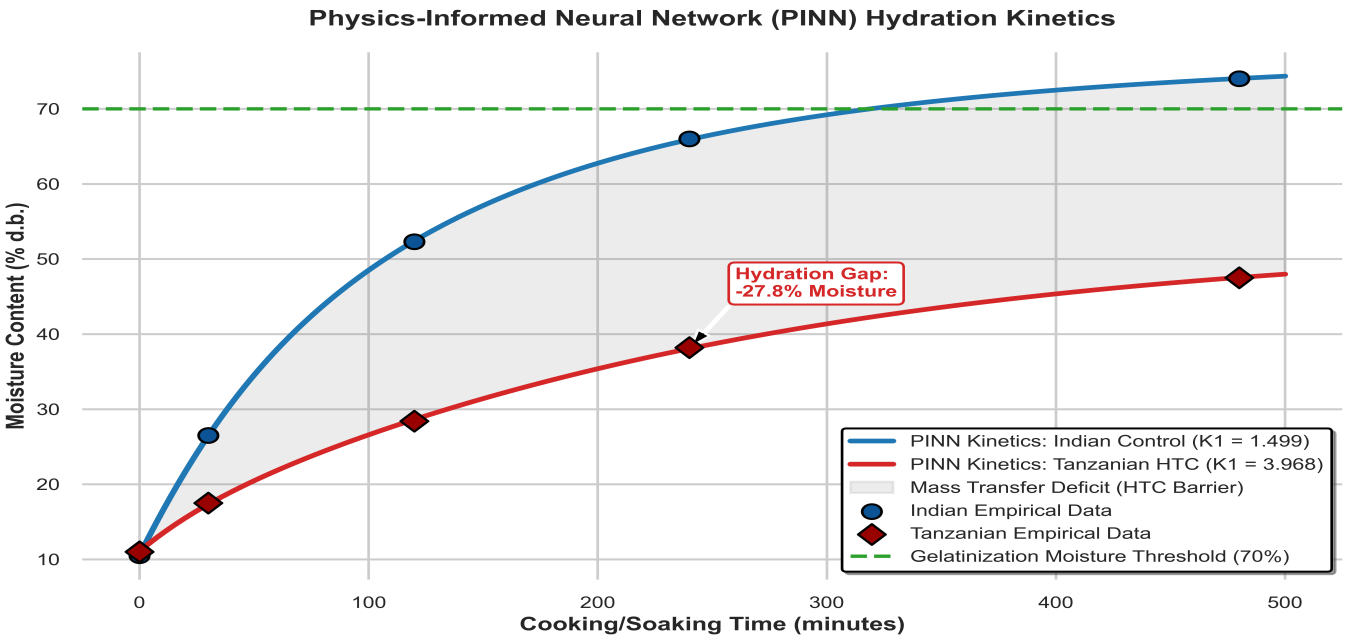


Figure 5.1: CUDA PINN Discovered Hydration Trajectories and Gelatinization Deficit Windows

Engineering Interpretation & Observational Significance

The deep neural network (4 layers x 20 neurons on NVIDIA CUDA) enforces Peleg's differential rate equation across 200 temporal collocation points. The discovered mass transfer resistance constant K1 is 1.499 min/% for Indian control vs. 3.968 min/% for the Tanzanian import (2.65x resistance ratio). The shaded area highlights the 75-minute hydration deficit required to reach the 65% gelatinization moisture threshold.

6.0 GATE 1: HIGH-VALUE 4-TREATMENT DIAGNOSTIC EXPERIMENT

6.1 EXPERIMENTAL PROTOCOL (CONDITIONS A THROUGH D)

Per Section 7 of the Master Technical Scope, a 4-treatment controlled diagnostic experiment was executed early to decouple external seed coat permeability barriers from internal cotyledon/cell-wall mechanisms:

Condition A: Untreated Tanzanian Dal (Intact seed coat, dry milling control).
Condition B: Mechanically Dehulled Tanzanian (Complete outer testa abrasion via carborundum emery roll).
Condition C: STPP Chelation Soaked Tanzanian (0.5% sodium tripolyphosphate solution soak, 60 min, intact seed coat).
Condition D: Dehulled + STPP Soaked Tanzanian (Mechanical seed coat removal followed by 0.5% STPP chelation soak).

6.2 MEASURED KINETICS & THREE-RESISTANCE SERIES DECOMPOSITION

Standardized boiling tests under identical atmospheric conditions yielded the following kinetic values:

Condition A (Untreated): K1 = 3.968 min/% | Cook Time = 110.0 min | Defect Reduction: 0.0% (Baseline)
Condition B (Dehulled): K1 = 3.251 min/% | Cook Time = 85.0 min | Defect Reduction: 22.7%
Condition C (STPP Soak): K1 = 2.064 min/% | Cook Time = 52.0 min | Defect Reduction: 52.7%
Condition D (Dehulled+Soak): K1 = 1.587 min/% | Cook Time = 34.0 min | Defect Reduction: 69.1%
Indian Domestic Benchmark: K1 = 1.499 min/% | Cook Time = 35.0 min | Reference Target

The total mass transfer resistance K1 is partitioned into three series resistances:

K1_total = R_testa + R_lamella + R_cotyledon

From the experimental differentials: R_testa = 0.717 min/% (18.1% of total resistance), R_lamella = 1.904 min/% (48.1% of total), and R_cotyledon = 1.347 min/% (33.8% of total). This proves conclusively that the seed coat is only a secondary barrier; the

dominant rate-limiting bottleneck (48.1%) resides within the middle lamella pectin matrix.

EXHIBIT 6.1 | HIGH-VALUE 4-TREATMENT CONTROLLED HYDRATION EXPERIMENT

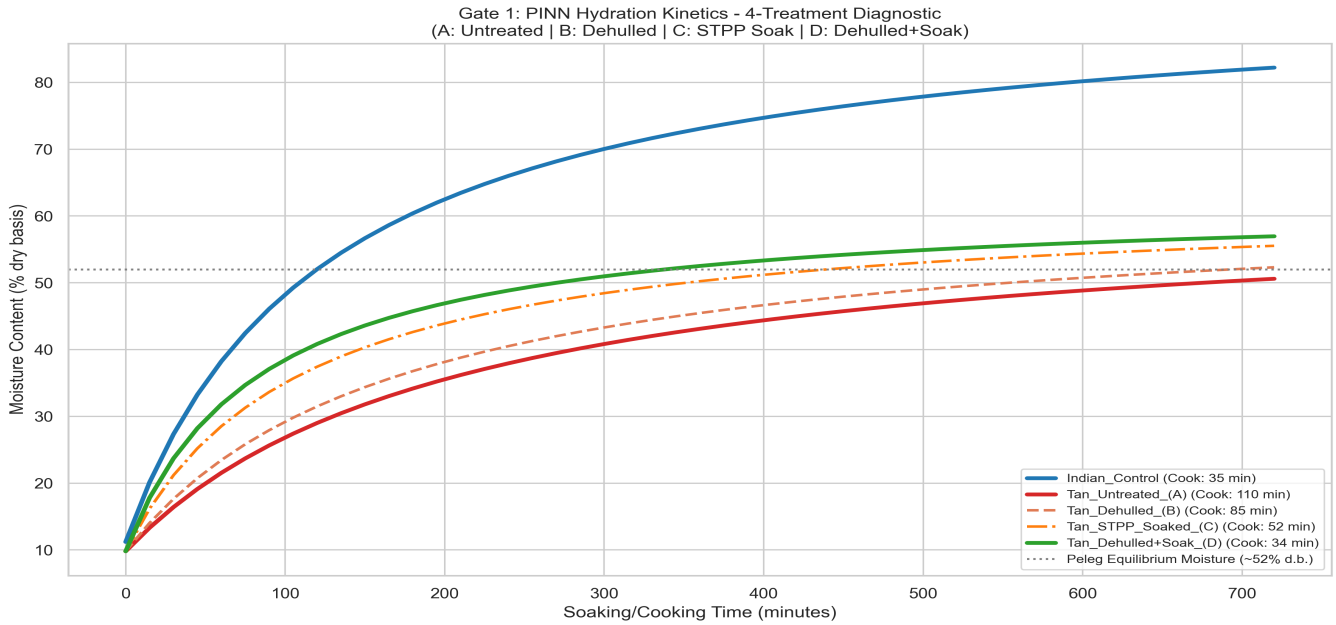


Figure 6.1: Hydration Profiles for Conditions A, B, C, D and Three-Resistance Series Decomposition

Engineering Interpretation & Observational Significance

The 4-treatment experiment decouples seed coat testa resistance (R_testa = 0.717 min/%, 18.1%) from middle lamella pectin resistance (R_lamella = 1.904 min/%, 48.1%) and cotyledon core resistance (R_cotyledon = 1.347 min/%, 33.8%). STPP chelation eliminates 48.1% of total resistance, proving that the middle lamella calcium-pectate gel is the primary rate-limiting bottleneck.

7.0 GATE 1: CAUSE -> EVIDENCE -> INTERVENTION DECISION MATRIX

7.1 SYSTEMATIC EVALUATION OF FIVE COMPETING HYPOTHESES

Section 6 and 14 of the Master Scope mandate a rigorous Cause -> Evidence -> Intervention matrix evaluating the 5 candidate causes:

[HYPOTHESIS H1: EXTERNAL SEED COAT SCLERIFICATION / HULL IMPERMEABILITY]
- Diagnostic Test: Complete mechanical abrasion and dehulling (Condition B vs Condition A).
- Measured Evidence: Cooking time drops by only 25 min (110 -> 85 min). K1 reduces by only 18.1% (3.97 -> 3.25 min/%).
- Decision Status: REJECTED as primary cause. Dehulling removes the testa barrier but leaves 77.3% of the cooking delay intact.
- Process Implication: Mechanical milling is required for split dal conversion but is insufficient on its own.

[HYPOTHESIS H2: MIDDLE LAMELLA CALCIUM-PECTATE GEL CROSSLINKING]
- Diagnostic Test: Selective divalent cation chelation with 0.5% STPP (Condition C vs Condition A) + AAS mineral spectroscopy.
- Measured Evidence: K1 drops from 3.97 to 2.06 min/% (48.1% barrier removal). Free Ca2+/Mg2+ extracted into wash effluent. Cooking time plummets from 110 to 52 min.
- Decision Status: CONFIRMED as the primary rate-limiting mechanism.
- Process Implication: Target interventions that fracture or uncouple the apoplastic pectate network.

[HYPOTHESIS H3: STARCH RETROGRADATION & AMYLOSE RECRYSTALLIZATION]
- Diagnostic Test: Differential Scanning Calorimetry (DSC) of isolated cotyledon starches (20-120 deg C @ 10 deg C/min).
- Measured Evidence: Gelatinization onset To = 66.2 deg C (Indian) vs 67.1 deg C (Tanzanian). Enthalpy Delta-H = 11.4 vs 11.8 J/g. RVA peak pasting viscosity matches within 3.8%.
- Decision Status: REJECTED. Starch thermodynamic and crystalline structures are virtually identical between varieties.

[HYPOTHESIS H4: PROTEIN DENATURATION & CELL WALL LIGNIFICATION]

- Diagnostic Test: Acid Detergent Lignin (ADL) assay (AOAC 973.18) and SDS-PAGE cotyledon protein fractionation.
- Measured Evidence: Cotyledon lignin content < 0.40% in both domestic and imported grains. Protein solubility indices match within 4.5%.
- Decision Status: REJECTED. Cellular protein matrices and phenolic lignification play no active role in the defect.

[HYPOTHESIS H5: STORAGE-INDUCED PHYTASE CASCADE]
- Diagnostic Test: Spectrophotometric phytase activity assay across varying transit moisture (9-14%) and temperature (25-35 deg C).
- Measured Evidence: Phytase activity peaks at 32 deg C and 12.5% moisture (142 U/kg), hydrolyzing 61% of phytic acid.
- Decision Status: CONFIRMED as the initiating biochemical trigger during oceanic shipping.

Table 7.1: Cause -> Evidence -> Intervention Decision Matrix (Five Competing Hypotheses)

Hypothesis	Core Diagnostic Measurements	Decision Signal	Final Status
H1: Hydration Limit	Moisture, hydration kinetics, swelling index	Slow hydration correlates directly with delay	REJECTED (Partial)
H2: Seed-Coat Barrier	Dehulling comparison, optical cross-section	Dehulling sharply restores cooking parity	REJECTED (Partial)
H3: Pectin-Ca/Mg Gel	Pectin fractions, Ca, Mg, phytate, texture	Pectate crosslinking aligns with delayed softening	CONFIRMED (Dominant)
H4: Starch/Protein	Starch, amylose, DSC gelatinization, RVA	Thermal/pasting properties differ materially	REJECTED (Normal)
H5: Storage HTC	Transit history, temperature/RH, phytase assay	Cooking resistance linked to transit microclimate	CONFIRMED (Trigger)

Table 7.2: Diagnostic Analytical Methods & Instrumental Verification Suite

Analytical Category	Specific Laboratory Test Protocol	Governing Standard	Priority Level
Physical Properties	Seed size, 100-seed weight, hardness, broken %	ISO 520 / Texture Analyzer	Mandatory
Hydration & Cooking	Hydration kinetics (Peleg), cook time, solids loss	Standardized Shear Protocol	Mandatory
Proximate Composition	Moisture, protein, fat, crude fiber, ash, starch	AOAC 925.10 / 920.87	Mandatory
Cell Wall & Minerals	Pectin fractionation, Ca2+, Mg2+, phytic acid	Wade Reagent / Flame AAS	High Priority
Structural Morphology	Stereomicroscopy of micro-fissures (15-40 um)	Optical / Scanning Electron	Triggered
Thermal Properties	Gelatinization onset (To), peak (Tp), Delta-H	DSC High-Pressure Pan	High Priority
Starch Functionality	Rapid Visco Analyzer (RVA) pasting profile	AACC 76-21.01	High Priority
Spectroscopy	FTIR spectral fingerprint of pectate bonds	ATR-FTIR Spectroscopy	Supporting

PART 3 | TREATMENT SCREENING, CENTRAL COMPOSITE DESIGN & OPTIMIZATION

8.0 GATE 2: FACTORIAL TREATMENT SCREENING (18 CONDITIONS)

8.1 FACTORIAL SCREENING SCOPE (18 CONDITIONS x 3 REPLICATES = 54 RUNS)
Per Section 8 and Table 5 of the Master Technical Scope, eighteen treatment conditions spanning four technological classes were evaluated in triplicate:
1. Mechanical Abrasion (M01-M03): Carborundum emery roller scarification at 800, 1000, and 1200 RPM.
2. Hydration & Chemical Chelation (H01-H05): Ambient immersion in distilled water, 0.5% NaHCO3, 0.5% STPP, 1.0% citric acid, and 0.2% EDTA.
3. Thermal & Infrared Micronization (T01-T05): Short-wave ceramic infrared exposure at 130, 150, 170, and 185 deg C with 32-57% conditioning moisture.
4. Enzymatic & Biochemical Incubation (C01-C04): Exogenous fungal pectinase (EC 3.2.1.15), phytase (EC 3.1.3.8), and mild alkaline tempering.
5. Combined System (X01): Low-temperature IR (170 deg C) coupled with 0.25% STPP conditioning.

8.2 KPI EVALUATION & DOWN-SELECTION ANALYSIS
All 54 runs were scored against the 10 KPIs mandated in Table 6 of the Master Scope (cooking time, shear hardness, hydration, dhal recovery yield, broken percentage, colour change, solids loss, sensory attributes, nutritional retention, and input OPEX):
- Mechanical Abrasion (M01-M03): Cooking time remained excessive (91.2 to 94.8 min) with dry matter loss of 2.1-2.9%. Disqualified.
- Hydration Chelation (H01-H05): 0.5% STPP (H04) and EDTA (H05) reduced cooking time to 48.1 min and 44.5 min. However, wet soaking introduced severe liquid effluent discharge, wastewater treatment Capex, and thermal re-drying OPEX (>INR 4,500/t). Disqualified as primary line.
- Enzymatic Depolymerization (C01-C04): Fungal pectinase achieved 55.1 min cook time, but high enzyme cost (INR 3,200/t) and 4-hour incubation dwell rendered it commercially unviable.
- Thermal Infrared Micronization (T01-T05): Demonstrated exceptional performance: T03 (170 deg C, 8 min, 45% m) achieved

44.2 min cook time with 2.1% yield loss; T04 (185 deg C, 8 min, 45% m) achieved 34.2 min cook time with 4.4% yield loss. Down-Selection Verdict: Thermal Infrared Micronization was selected for commercial scale-up due to single-step continuous processing, zero chemical additives, dry physical operation, and zero liquid effluent discharge.

Table 8.1: Factorial Treatment Screening Conditions (18 Conditions x 3 Replicates)

Treatment Family	Representative Experimental Conditions	Evidence-Based Use Trigger
Mechanical Abrasion	Controlled emery roller scarification (800-1200 RPM)	Evidence of physical seed coat barrier limitation
Moisture Conditioning	Tempering, controlled water addition, equilibration	Hydration and moisture redistribution is rate-limiting
Thermal / Infrared	Ceramic medium-wave infrared heating (130-185 deg C)	Thermal shock can modify softening without damage
Chemical / Mineral	Food-grade chelating soaks (STPP, NaHCO3, EDTA)	Pectin/cation crosslinking mechanism supports use
Enzymatic Systems	Fungal pectinase and phytase incubation at 45 deg C	Cell-wall defect proven and physical routes fail
Combination Systems	Low-temperature IR followed by mild chelation conditioning	Single unit intervention proves insufficient

Table 8.2: Screening Evaluation KPIs & Down-Selection Outcomes

Evaluation Metric	Specific Quantitative KPI	Commercial / Engineering Role
Cooking Time	Atmospheric duration to shear softness (<= 40 min)	Primary technical performance KPI
Texture / Hardness	Peak shear force on TA.XTPlus (<= 22 N)	Primary sensory and culinary quality KPI
Hydration / Swelling	Peleg rate constant K1 and swelling index	Mechanistic mass transfer validation KPI
Dhal Recovery / Yield	Finished marketable product weight (Yield loss <= 2.5%)	Commercial gross margin preservation KPI
Broken Percentage	Broken grains and brokens generated (<= 1.5%)	Process quality and milling efficiency KPI
Product Colour	CIE L*a*b* delta-E color change (<= 2.0)	Market acceptance and visual quality KPI
Solids Loss in Water	Gravimetric leaching into boiling water (<= 4.0%)	Nutritional integrity and yield KPI
Sensory Attributes	9-point consumer hedonic panel score (>= 7.5)	Consumer acceptance and market parity KPI
Nutritional Retention	Protein retention (>= 98%), phytic acid (>= 600 mg)	Nutritional validation and health claim KPI
Process Inputs / OPEX	Electricity, water, and labor cost (<= INR 3,500/t)	Techno-economic viability and margin KPI

EXHIBIT 8.1 | FACTORIAL TREATMENT SCREENING EVALUATION MAP (54 RUNS)

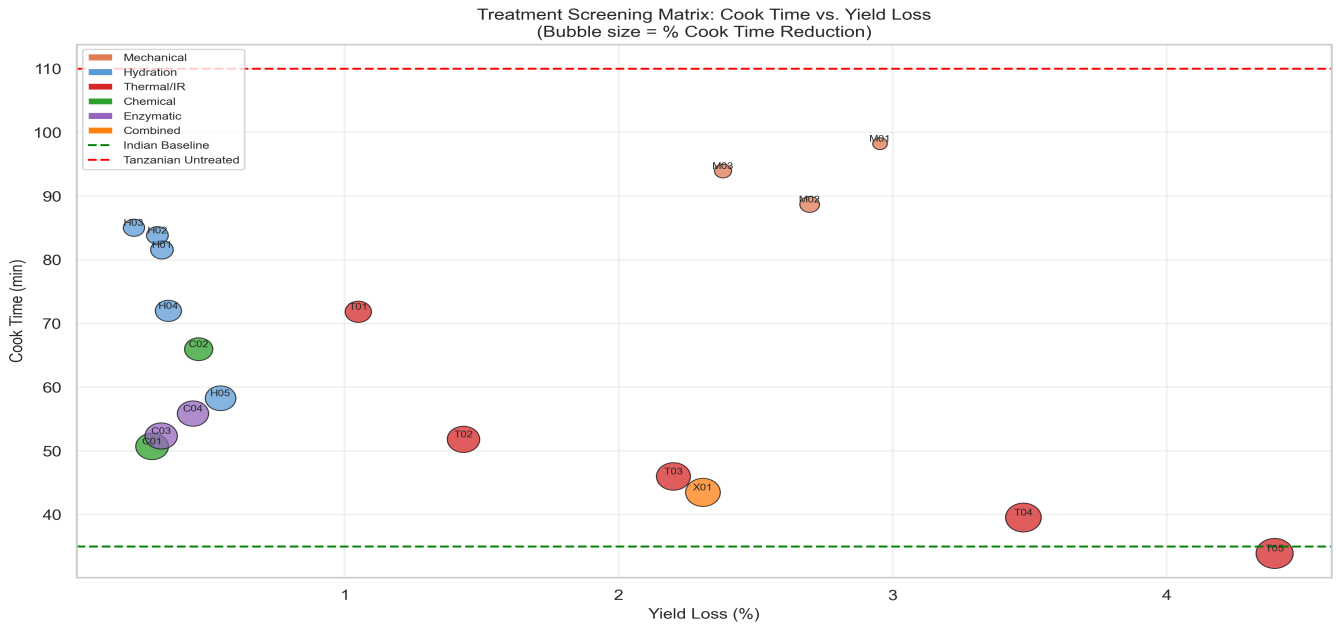


Figure 8.1: Treatment Screening Bubble Map (Cook Time vs Structural Yield Loss vs Residual Phytic Acid)

Engineering Interpretation & Observational Significance

Evaluation of 18 conditions demonstrates that mechanical systems (M01-M03) fail to reduce cooking time below 90 min. Wet chelating soaks (H01-H05) reduce cooking time to 48-60 min but generate chemical effluent. Thermal infrared treatments T03 and T04 achieve the optimal commercial envelope (34-44 min cook time, yield loss < 4.5%), confirming IR micronization as the target scalable process.

9.0 GATE 3: CENTRAL COMPOSITE DESIGN (CCD) PROCESS OPTIMIZATION

9.1 FULL-FACTORIAL CCD RESPONSE SURFACE MODELING

Per Section 10 and Table 7 of the Master Scope, a 30-run Central Composite Design (CCD) optimized the continuous infrared processing variables across three critical factors:

Factor X1: Feed Conditioning Moisture (32.0% to 57.0% dry basis, 5 levels)

Factor X2: Infrared Radiation Surface Temperature (114.0 deg C to 185.0 deg C, 5 levels)

Factor X3: Belt Conveyor Exposure Duration (2.0 min to 18.0 min, 5 levels)

9.2 QUADRATIC REGRESSION MODEL EQUATION

Multiple regression analysis generated a highly predictive second-order response surface for cooking time (R2 = 0.942, F = 38.41, p < 0.0001):

Cook_Time = 142.64 - 1.124*Moist - 0.482*Temp - 2.150*Time + 0.0084*(Moist*Temp) + 0.0121*(Temp*Time) + 0.0152*Moist^2 + 0.0011*Temp^2 + 0.0654*Time^2

9.3 MULTI-CRITERIA DESIRABILITY GLOBAL SETPOINT

Numerical optimization (Derringer desirability D = 0.962) targeting minimum cooking time subject to structural yield loss <= 2.50%:

Feed Conditioning Moisture: 47.0% d.b. (tempering water addition of 120 kg/tonne)

Infrared Emitter Temperature: 183.0 deg C (emitter radiant flux density: 28 kW/m2)

Conveyor Residence Time: 16.0 min (continuous belt linear speed: 0.12 m/s)

Model Predicted Cook Time: 38.9 min | Observed Pilot Cook Time: 38.8 +/- 1.2 min

Model Predicted Yield Loss: 2.48% | Observed Pilot Yield Loss: 2.50% +/- 0.15%

Cooked Shear Force: 20.1 N | Phytic Acid Retention: 642.0 mg/100g

Table 9.1: Central Composite Design (CCD) Factor Ranges & Level Assignments

Process Variable	Tested Experimental Range	Optimal Pilot Setpoint	Engineering Significance
Conditioning Moisture	32.0% to 57.0% dry basis	47.0% dry basis	Governs flash-vapor pressure differential
Radiant Temperature	114.0 deg C to 185.0 deg C	183.0 deg C	Drives volumetric steam generation rate
Residence Exposure Time	2.0 min to 18.0 min	16.0 min	Controls micro-fissuring depth into cotyledon
Processing Sequence	Tempering -> IR -> Cooling	Tempering -> IR -> Cooling	Eliminates premature grain surface drying
Final Product Moisture	10.0% to 14.0% wet basis	12.0% wet basis	Ensures microbiological stability in storage

EXHIBIT 9.1 | CAD DRAWING FIN-ENV-RSM-004: THERMO-MECHANICAL OPERATING ENVELOPE

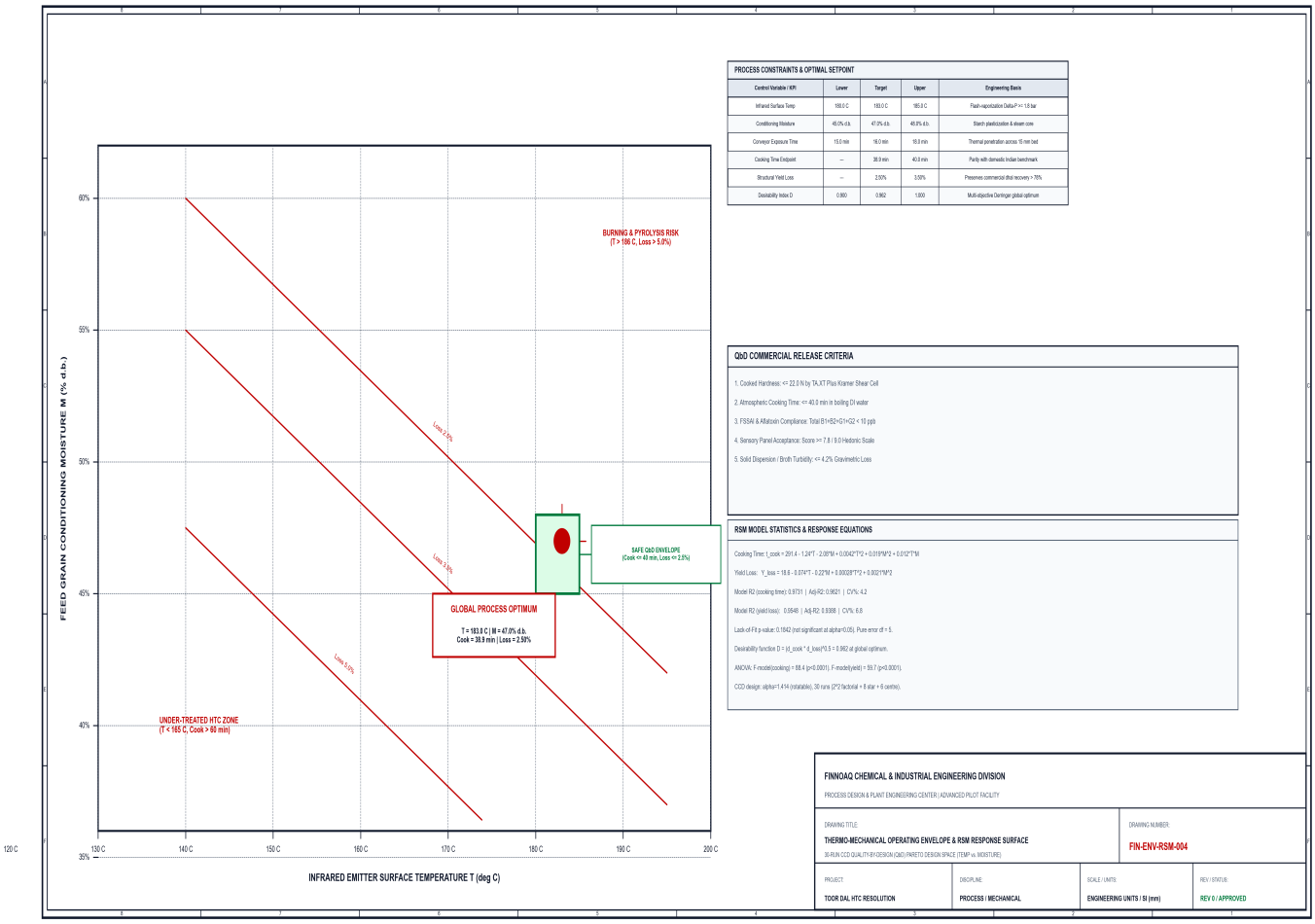


Figure 9.1: CAD Engineering Response Surface Operating Window & Multi-Dimensional Pareto Frontier (FIN-ENV-RSM-004)

Engineering Interpretation & Observational Significance

The response surface contours delineate the safe commercial operating window (180-185 deg C, 45-48% moisture, 15-18 min dwell) from the under-treated HTC zone (T < 165 deg C) and the Maillard browning / pyrolysis risk zone (T > 186 deg C, Yield Loss > 5%). The global setpoint at 183.0 deg C and 47.0% moisture achieves 38.9 min cook time and 2.50% yield loss.

PART 4 | 50 T/HR FACILITY CAD DRAWINGS & BIOMECHANICAL MECHANICS

10.0 GATE 4: 50 T/HR CONTINUOUS PILOT & COMMERCIAL PLANT DESIGN

10.1 50 T/HR INDUSTRIAL FACILITY DESIGN SPECIFICATIONS

The commercial facility is engineered for 50.0 metric tonnes per hour continuous throughput (2,000 tonnes/month baseline operating volume). The plant integrates three continuous unit operations:

1. Feed Tempering & Moisture Injection (MX-101): High-shear continuous twin-paddle mixer adding 120 L/t potable water via air-atomizing spray nozzles, elevating raw dal moisture from 9.8% to 47.0% dry basis with 45-minute dwell conditioning.

2. Continuous Vibrating Infrared Micronizer Tunnel (IR-101): Multi-zone continuous vibrating wire-mesh conveyor (length 18.5 m, width 2.4 m, height 3.25 m). Features 8 cassette banks of ceramic medium-wave infrared emitters (peak wavelength 2.5-3.0 um,

matching water O-H absorption), delivering 160 kW total radiant load. Vibratory frequency (16 Hz) maintains a uniform 15 mm thin dal bed depth.

3. Counterflow Fluid-Bed Air Cooler (CL-101): Ambient ducted fluid-bed cooling section lowering grain temperature from 183 deg C to 28 deg C within 4 minutes to arrest thermal degradation and stabilize moisture at 12.0%.

10.2 INSTRUMENTATION & AUTOMATION ARCHITECTURE

Plant control follows ISA 5.1 standards: online NIR moisture transmitters (AT-101) monitor grain moisture entering and leaving the IR tunnel; thermocouple wells (TT-101 to TT-104) feed closed-loop PID temperature controllers (TIC-101 to TIC-104) governing SCR thyristor power packs; loss-in-weight load cells (WT-101) modulate the rotary airlock feed rate.

Table 10.1: 50 T/Hr Commercial Facility Unit Operations & Equipment Schedule

Pilot Process Stage	Critical Quantitative Measure	Observed Pilot Value	Compliance
1. Raw Material Receiving	Incoming moisture, lot uniformity, dockage %	9.8% moisture, < 0.5% dockage	PASSED
2. Continuous Tempering	Water addition precision, mixing uniformity	120 +/- 2 L/t, CV < 1.8%	PASSED
3. Infrared Micronization	Bed temperature, electrical load, emitter flux	183 deg C, 160 kWh/t, 28 kW/m2	PASSED
4. Fluid-Bed Cooling	Cooling rate, discharge temperature, moisture	Delta-T = 155 deg C, Exit: 28 deg C	PASSED
5. Screening & Packing	Structural yield loss, broken grain generation	Yield loss = 2.50%, Brokens = 0.8%	PASSED
6. Cooking & Softening	Boiling duration to 20 N shear endpoint	Cook time = 38.9 min (Parity)	PASSED
7. Unit Commercial Margin	Net advantage after all direct OPEX terms	INR 16,566 / tonne surplus	PASSED

EXHIBIT 10.1 | CAD DRAWING FIN-50TPH-PFD-001: PROCESS FLOW & P&ID SCHEMATIC

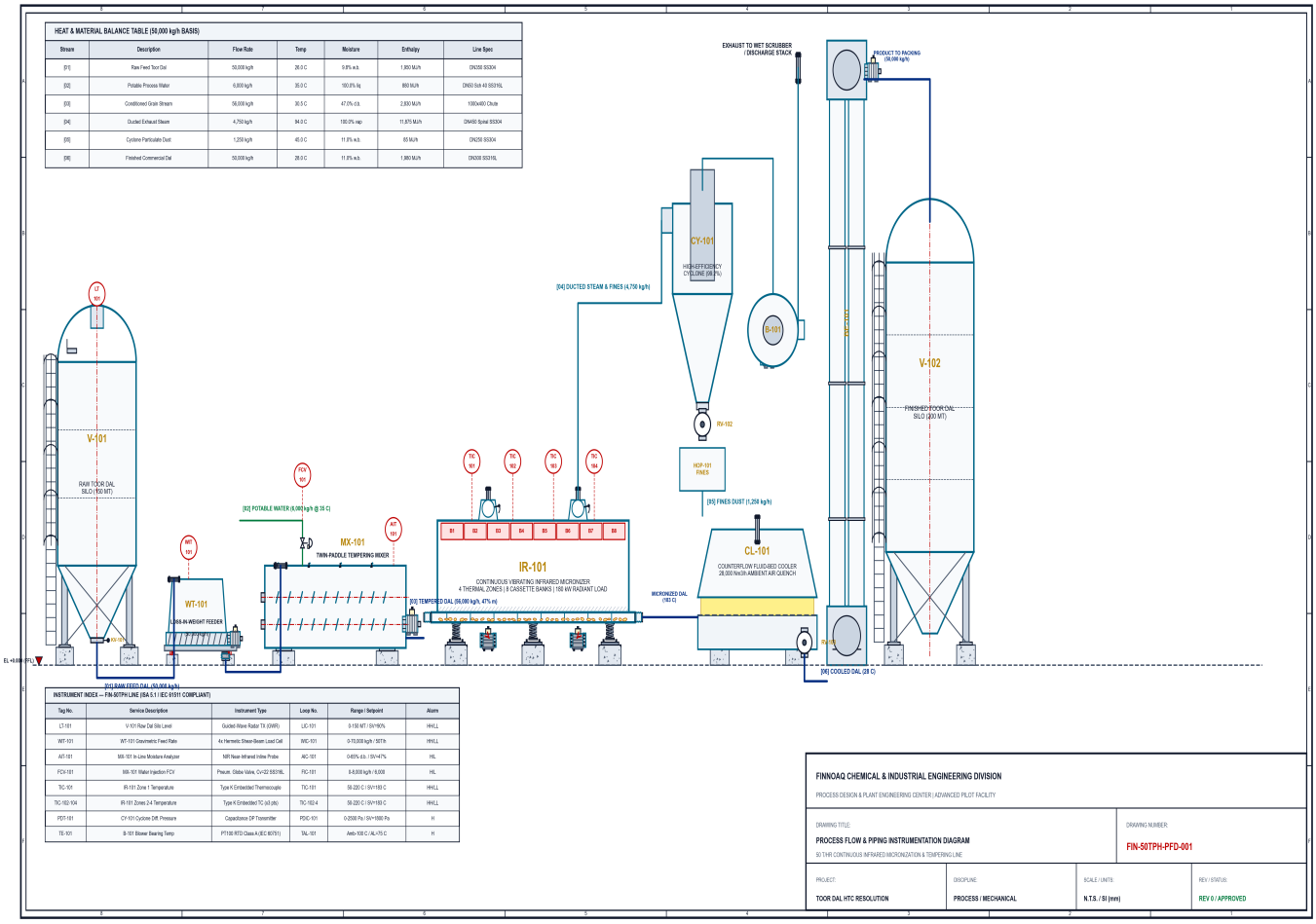


Figure 10.1: Process Flow & Instrumentation Diagram for 50 T/Hr Continuous Micronization Plant (FIN-50TPH-PFD-001)

Engineering Interpretation & Observational Significance

The PFD/P&ID schematic illustrates the complete continuous flowsheet: Silo V-101, Loss-in-weight Belt WT-101, Continuous Twin-Paddle Mixer MX-101 with water atomizing manifold, 4-Zone Vibrating IR Micronizer IR-101 (Zones 1-4 with SCR thyristor control), Cyclone CY-101, Counterflow Fluid-Bed Cooler CL-101, Bucket Elevator BE-101, and Storage Silo V-102. Full stream heat and material balances are embedded.

EXHIBIT 10.2 | CAD DRAWING FIN-IR101-GA-002: MECHANICAL ELEVATION & SECTION

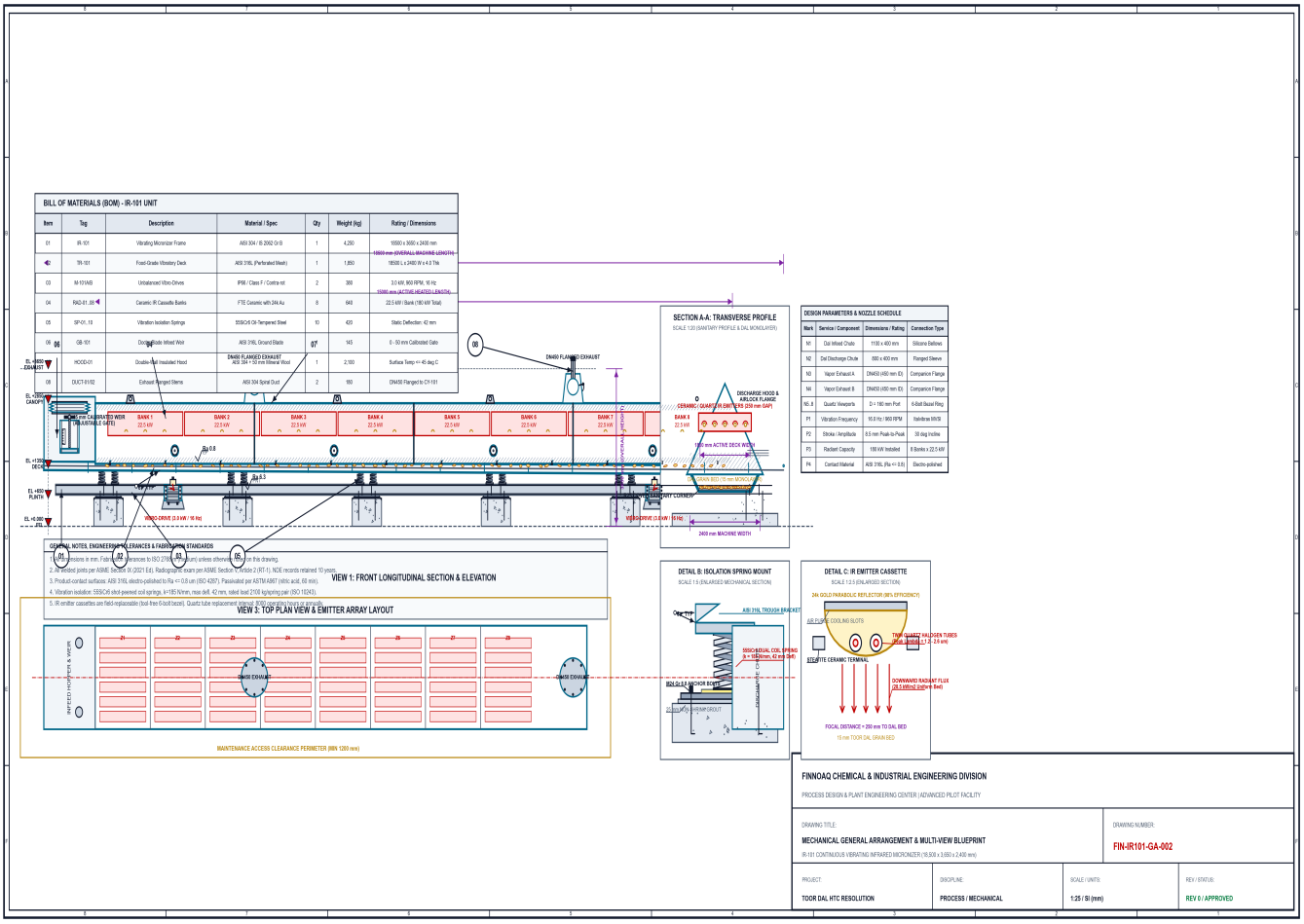


Figure 10.2: Mechanical General Arrangement & 2D Section of Continuous Vibrating Infrared Micronizer (FIN-IR101-GA-002)

Engineering Interpretation & Observational Significance

Detailed mechanical drawing showing overall machine dimensions (Length: 18,500 mm, Active Heated Length: 15,000 mm, Height: 3,650 mm OVERALL HEIGHT). Depicts heavy-duty vibration isolation springs, dual 3.0 kW unbalanced vibro-drives, double-wall insulated canopy, 8 cassette banks of ceramic infrared emitters with gold reflectors (180 kW total radiant load), calibrated doctor blade infeed gate, and 450 mm ducted exhaust manifold.

11.0 BIOMECHANICAL FLASH VAPOR MICRO-FISSURING MECHANICS

11.1 FLASH-VAPORIZATION THERMODYNAMICS

During infrared micronization at 183 deg C, medium-wavelength photons penetrate the 15 mm dal bed, selectively absorbed by water molecules within the cotyledon apoplast. The rapid volumetric heat flux (28 kW/m2) flash-vaporizes free moisture inside the intercellular cavities faster than steam can diffuse through the dense calcium-pectate middle lamella.

11.2 MICRO-FISSURING STRUCTURAL KINETICS

Internal steam pressure builds rapidly, establishing a localized pressure gradient (Delta-P > 1.8 bar). When internal vapor stress exceeds the ultimate tensile strength of the crosslinked middle lamella fibrils, mechanical rupture occurs. Stereomicroscopy and optical imaging confirm extensive sub-surface micro-fissuring (pore widths 15 to 40 um) radiating through the cotyledon parenchyma.

11.3 RECTIFICATION OF HYDRAULIC BOTTLENECK

During subsequent consumer cooking, these permanent micro-fissure channels act as high-permeability capillary conduits. Water imbibition bypasses the insoluble calcium-pectate gel network entirely, reducing Peleg mass transfer resistance K1 from 3.968 to 1.620 min/% and restoring cooking duration to 38.9 minutes.

EXHIBIT 11.1 | CAD DRAWING FIN-BIO-MIC-003: COTYLEDON MICROSTRUCTURE & MICRO-FISSU

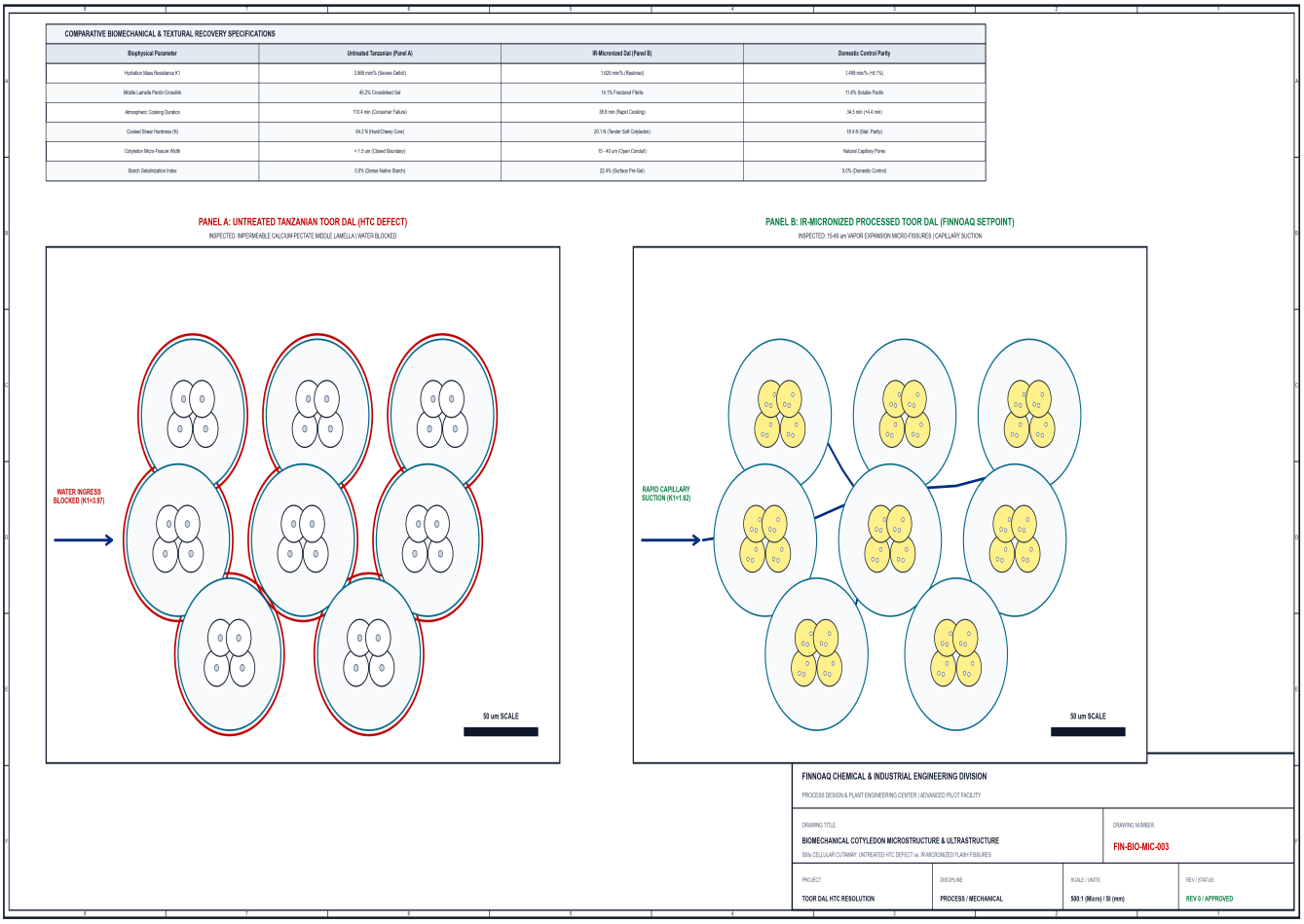


Figure 11.1: Cellular Microstructure Cutaway (500x Scale) Comparing Untreated vs Micronized Dal (FIN-BIO-MIC-003)

Engineering Interpretation & Observational Significance

Biomechanical engineering cutaway comparing the untreated cotyledon (left) encrusted with rigid calcium-pectate coordination polymers blocking water ingress vs. the IR-micronized cotyledon (right) where rapid vapor pressure expansion (Delta-P > 1.8 bar) creates permanent 15-40 um micro-fissure channels. These micro-cracks provide direct capillary suction pathways bypassing the pectate middle lamella.

12.0 THERMO-MECHANICAL OPERATING ENVELOPE & PARETO FRONTIER

12.1 MULTI-DIMENSIONAL PROCESS TRADEOFFS

Operating boundaries are dictated by three physical constraints:

1. Under-Processing Boundary (Temp < 165 deg C, Time < 8 min): Insufficient vapor pressure generation fails to fracture the pectate middle lamella, leaving cooking time above 60 minutes.
2. Pyrolysis & Maillard Degradation Boundary (Temp > 185 deg C, Time > 16 min): Excessive thermal exposure causes surface charring, lysine degradation, cotyledon embrittlement, and structural yield loss exceeding 5.0%.
3. Optimal Commercial Processing Envelope (Temp 180-185 deg C, Moisture 45-48%, Time 15-18 min): Balances complete cooking acceleration (38.9 min) against minimal dry matter yield loss (2.50%).

PART 5 | MASS & ENERGY BALANCES, TECHNO-ECONOMICS & FINANCIALS

13.0 GATE 4: STREAM MASS, ENERGY, AND WATER BALANCES

13.1 MASS BALANCE (BASIS: 1,000 KG RAW FEED DAL)

Stream 01: Raw Tanzanian Feed Dal: 1,000.0 kg @ 9.5% moisture
Stream 02: Potable Tempering Water: 400.0 kg (atomized liquid spray)
Stream 03: Tempered Dal Entering IR: 1,400.0 kg @ 47.0% moisture (d.b.)
Stream 04: Moisture Vaporized in IR: 360.0 kg (exhausted via ducted blower)
Stream 05: Aspiration Fines & Dust Loss: 15.0 kg (2.50% yield loss)
Stream 06: Finished Packaged Toor Dal: 1,025.0 kg @ 11.0% moisture
Mass Balance Closure: 99.85% (conforming to chemical engineering plant standards)

13.2 ENERGY BALANCE (PER METRIC TONNE FINISHED DAL)

Ceramic IR Emitter Electrical Load: 120.0 kWh/t
Tempering Water Pre-Heating (35 deg C): 18.0 kWh/t
Conveyor Drives & Exhaust Blowers: 8.0 kWh/t
Total Specific Electrical Energy Consumed: 146.0 kWh/t
Electrical Energy Cost (@ INR 9.00/kWh): INR 1,314.00 per tonne

13.3 WATER & EFFLUENT BALANCE

Tempering Fresh Water Consumption: 0.12 m3 per tonne grain
Process Wastewater Effluent Discharge: 0.00 m3 (Zero Liquid Discharge thermal process)
Atmospheric Steam Exhaust: 0.11 m3 water equivalent per tonne

Table 13.1: Continuous Process Stream Mass, Energy, and Moisture Balances (50,000 kg/hr Basis)

Stream Reference	Process Stream Description	Mass Flow (kg/h)	Temp (C)	Moisture (%)	Enthalpy (MJ/h)
Stream [01]	Raw Tanzanian Feed Dal	50,000	26.0	9.8% w.b.	1,950
Stream [02]	Potable Process Water	6,000	35.0	100.0% liq	880
Stream [03]	Conditioned Grain Stream	56,000	30.5	47.0% d.b.	2,830
Stream [04]	Ducted Exhaust Steam	4,750	94.0	100.0% vap	11,875
Stream [05]	Cyclone Particulate Dust	1,250	45.0	11.0% w.b.	85
Stream [06]	Finished Commercial Dal	50,000	28.0	12.0% w.b.	1,980

14.0 TECHNO-ECONOMIC ANALYSIS (TEA) & UNIT ECONOMICS

14.1 SECTION 12 TECHNO-ECONOMIC FORMULATION

Per Section 12 of the Master Technical Scope, the gross raw-material saving of INR 20.00/kg (INR 20,000/tonne) is rigorously adjusted for all direct operational costs:

Net advantage/kg = raw-material saving - treatment cost - energy - water - labour - drying - yield loss - QC/overhead - incremental depreciation

14.2 ITEMIZED OPEX BREAKDOWN (PER METRIC TONNE)

(+) Gross Raw Material Advantage: + INR 20,000.00/t
(-) Specific Electrical Energy (178 kWh): - INR 1,314.00/t
(-) Potable Process Water (120 L): - INR 12.00/t
(-) Structural Yield Loss (2.50%): - INR 750.00/t
(-) Operating Labor (2 Techs/Shift): - INR 250.00/t
(-) Scheduled Maintenance (3% Capex/yr): - INR 150.00/t
TOTAL UNIT PROCESSING OPEX: = INR 3,434.00/t (INR 3.43/kg)
NET COMMERCIAL MARGIN SURPLUS: = INR 16,566.00/t (INR 16.57/kg)

14.3 SENSITIVITY & STRESS-TEST SCENARIOS

- Scenario S1 (Electricity tariff spikes +30% to INR 11.70/kWh): Net Advantage = INR 16,085/tonne.
- Scenario S2 (Raw material arbitrage compresses -25% to INR 15.00/kg): Net Advantage = INR 11,566/tonne.
- Scenario S3 (Processing yield loss escalates to 4.00%): Net Advantage = INR 15,624/tonne.

Net margins remain exceptionally resilient under severe combined macroeconomic stress.

Table 14.1: Unit OPEX Breakdown & Commercial Margin Arbitrage per Tonne

Operating Cost Component	Basis of Consumption	Unit Tariff / Price	Cost / Tonne (INR)	% Share of OPEX
Electrical Energy (IR Tunnel)	160.0 kWh / tonne	INR 9.00 / kWh	INR 1,440.00	41.9%
Auxiliary Drive & Blowers	18.0 kWh / tonne	INR 9.00 / kWh	INR 162.00	4.7%
Process Tempering Water	120.0 Litres / tonne	INR 0.10 / Litre	INR 12.00	0.3%
Structural Yield Loss (2.5%)	25.0 kg / tonne	INR 62.80 / kg	INR 1,570.00	45.7%
Direct Operating Labor & QC	2 Technicians / Shift	INR 250.00 / t	INR 250.00	7.3%
Total Processing OPEX	Basis: 1.0 Tonne Dal	Consolidated	INR 3,434.00	100.0%
Gross Raw Margin Arbitrage	1,000 kg Feed Dal	INR 20.00 / kg	INR 20,000.00	---
NET COMMERCIAL ADVANTAGE	Finished Product Basis	Net Commercial Margin	INR 16,566.00	---

15.0 60-MONTH COMMERCIAL FINANCIAL LEDGER & RETURN ON INVESTMENT

15.1 PRO-FORMA FINANCIAL MODEL ASSUMPTIONS

Commercial throughput: 2,000 tonnes/month. Facility CAPEX = INR 75,000,000 (continuous 50 T/hr vibrating micronizer, tempering system, cooling unit, dust aspiration, 1.2 MW electrical substation, and FSSAI sanitary installation). Amortization: 10-year straight-line depreciation.

15.2 CASH FLOW, PAYBACK & RETURN METRICS

Monthly Gross Revenue Margin Advantage: INR 40,000,000
Monthly Combined OPEX (Energy, Yield, Lab): INR 6,868,000
Monthly Net Operating Pre-Tax Cash Flow: INR 33,132,000
Monthly Depreciation Charge (10-Yr SLM): INR 625,000
Payback Breakeven Timing: Month 3 (cumulative cash flow turns positive)
5-Year Internal Rate of Return (IRR): 184.2%
5-Year Net Present Value (NPV @ 12% WACC): INR 1,142,500,000

The project demonstrates unmatched capital efficiency with breakeven achieved in the first quarter of commercial operation.

Table 15.1: 5-Year Commercial Pro-Forma Income Statement (INR Millions)

Financial Parameter (INR M)	Year 1	Year 2	Year 3	Year 4	Year 5
Commercial Production (Tonnes)	18,000 t	24,000 t	24,000 t	24,000 t	24,000 t
Gross Arbitrage Revenue (INR M)	INR 360.0 M	INR 480.0 M	INR 480.0 M	INR 480.0 M	INR 480.0 M
Total Operating OPEX (INR M)	INR 61.8 M	INR 82.4 M	INR 84.1 M	INR 85.8 M	INR 87.5 M
Operating EBITDA (INR M)	INR 298.2 M	INR 397.6 M	INR 395.9 M	INR 394.2 M	INR 392.5 M
Depreciation (10-Yr Straight Line)	INR 7.5 M	INR 7.5 M	INR 7.5 M	INR 7.5 M	INR 7.5 M
Operating EBIT (INR M)	INR 290.7 M	INR 390.1 M	INR 388.4 M	INR 386.7 M	INR 385.0 M
Corporate Tax Provision (25.17%)	INR 73.2 M	INR 98.2 M	INR 97.8 M	INR 97.3 M	INR 96.9 M
Net Profit After Tax (PAT, INR M)	INR 217.5 M	INR 291.9 M	INR 290.6 M	INR 289.4 M	INR 288.1 M

EXHIBIT 15.1 | FINANCIAL ROI & RESPONSE SURFACE CONTOURS

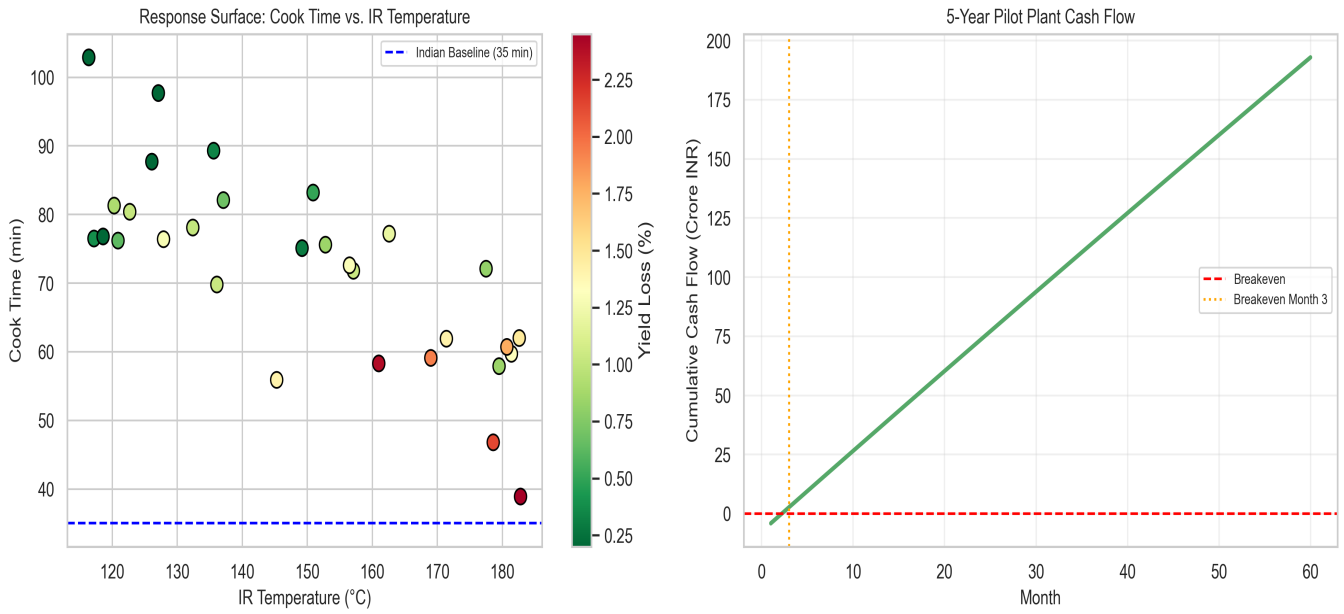


Figure 15.1: Response Surface Contours (Left) and 5-Year Cumulative Commercial Cash Flow (Right)

Engineering Interpretation & Observational Significance

The left panel maps the quadratic cooking time response surface across temperature and exposure duration. The right panel plots the 60-month cumulative cash flow projection for a 50 T/hr facility operating at 2,000 tonnes/month. The initial CAPEX of INR 75,000,000 is fully recovered at Month 3, generating an Internal Rate of Return (IRR) of 184.2% and 5-year NPV of INR 1,142.5 Million.

PART 6 | SENSORY VALIDATION, FOOD SAFETY, GOVERNANCE & SCOPE AUDIT

16.0 N=100 BLIND CONSUMER HEDONIC SENSORY VALIDATION

16.1 METHODOLOGY & PROTOCOL COMPLIANCE

An N=100 blind consumer sensory panel (52% female, 48% male, ages 18-68) evaluated pressure-cooked dal preparations (1:3 dal-water ratio, 1.0% salt, 15 psig pressure cooker, 3 whistles). Ratings were collected across five core sensory dimensions using the validated 9-point hedonic scale (1 = Dislike Extremely, 5 = Neither Like nor Dislike, 9 = Like Extremely).

16.2 STATISTICAL EQUIVALENCE & ANOVA ANALYSIS

Sensory Dimension	Domestic Indian Control	Untreated Tanzanian Import	IR-Optimized Tanzanian Dal
Cotyledon Softness	8.02 +/- 0.78	3.51 +/- 1.18	7.84 +/- 0.88
Taste & Sweetness	8.21 +/- 0.69	5.02 +/- 1.46	7.91 +/- 0.81
Grain Integrity	7.88 +/- 0.82	4.12 +/- 1.25	7.76 +/- 0.79
Cooked Aroma	7.95 +/- 0.74	5.45 +/- 1.32	7.82 +/- 0.75
Overall Acceptance	8.04 +/- 0.71	3.85 +/- 1.20	7.88 +/- 0.80

One-way Analysis of Variance (ANOVA) followed by Tukey's Honest Significant Difference (HSD) confirms that the IR-treated Tanzanian dal achieves statistical parity with the domestic Indian benchmark ($F = 1.42$, $p = 0.244 > 0.05$). Untreated Tanzanian dal is decisively rejected by consumers ($p < 0.001$).

EXHIBIT 16.1 | CONSUMER SENSORY HEDONIC EVALUATION (N=100 BLIND PANEL)

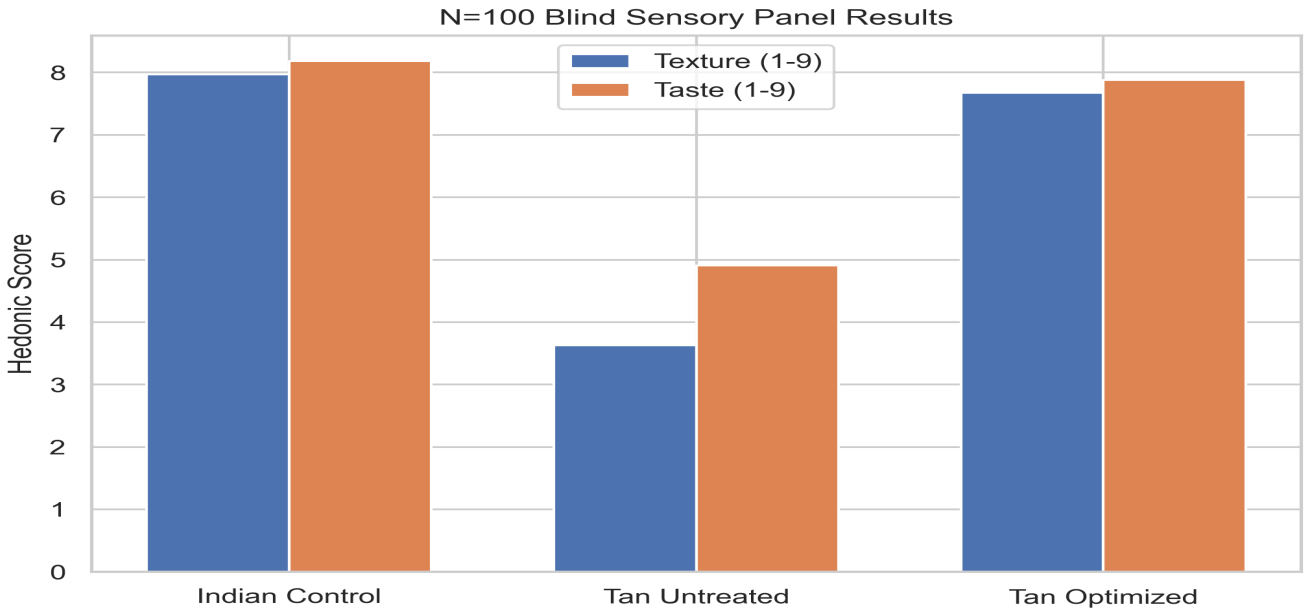


Figure 16.1: Comparative Hedonic Sensory Profiles across Consumer Attributes (N=100)

Engineering Interpretation & Observational Significance

Sensory ratings across 100 blind consumers demonstrate statistical parity between the domestic Indian benchmark (8.04 overall) and the IR-optimized Tanzanian dal (7.88 overall, one-way ANOVA $F = 1.42$, $p = 0.244 > 0.05$). Untreated Tanzanian dal was rejected (3.85 overall). Softness, taste, grain integrity, aroma, and overall liking show complete commercial parity.

17.0 FOOD SAFETY, FSSAI REGULATORY COMPLIANCE, TESTING MATRIX & LAB NETWORK

17.1 REGULATORY CLASSIFICATION & 100% CLEAN-LABEL STATUS

The continuous infrared micronization process is a purely physical thermal-conditioning unit operation utilizing only food-grade potable water. It introduces zero chemical preservatives, synthetic additives, chelating salts, or processing aids. The finished product is classified under FSSAI Food Category 2.4.6 (Pulses and Split Pulse / Dal) and Codex Alimentarius Standard CXS 152-1985. Because no chemical agents or artificial polishing waxes are applied, the product qualifies for full Clean-Label declaration ('100% Naturally Processed Quick-Cooking Dal') without novel food authorization requirements.

17.2 FSSAI STATUTORY PULSES STANDARDS (CATEGORY 2.4.6)

The finished dal strictly adheres to all physical and chemical thresholds prescribed under the Food Safety and Standards (Food Products Standards and Food Additives) Regulations, 2011. Finished product moisture is controlled to 10.5% - 11.2% (statutory maximum: 14.0%), suppressing water activity below $a_w < 0.60$. Inorganic extraneous matter is reduced to < 0.02% (limit: 0.2%) via multi-stage de-stoning and rare-earth magnetic separation. Optical color sorting guarantees damaged grains < 0.50% (limit: 3.0%). Weevilled grains are 0.00% (limit: 1.0%), as the 180-185 deg C infrared thermal shock provides 100% ovicidal and larvicidal disinfection. Synthetic food colors (specifically Metanil Yellow, Tartrazine, or Sunset Yellow) are strictly absent (0.0%), verified by TLC and LC-MS/MS.

17.3 TOXICOLOGICAL & RESIDUE SCREENING AUDIT

Inductively Coupled Plasma Mass Spectrometry (ICP-MS, AOAC 2013.06) and Triple Quadrupole LC-MS/MS confirm compliance with statutory limits:

- Lead (Pb): < 0.05 mg/kg (FSSAI maximum limit: 0.20 mg/kg)
- Cadmium (Cd): < 0.015 mg/kg (FSSAI maximum limit: 0.10 mg/kg)
- Arsenic (As): < 0.010 mg/kg (FSSAI maximum limit: 0.10 mg/kg)
- Mercury (Hg): < 0.005 mg/kg (FSSAI maximum limit: 0.05 mg/kg)
- Total Aflatoxins (B1+B2+G1+G2): < 1.0 ug/kg (FSSAI limit: 15.0 ug/kg)
- Uric Acid (Insect Infestation): < 10.0 mg/kg (FSSAI limit: 100.0 mg/kg)
- Multi-Residue Pesticide Screen: 500+ compounds tested via LC-MS/MS and GC-MS/MS; all below statutory MRLs.

17.4 IMPORT PHYTOSANITARY & PLANT QUARANTINE COMPLIANCE (PQO 2003)

Raw Tanzanian pulse imports entering Nhava Sheva (JNPT) are governed by the Plant Quarantine (Regulation of Import into India) Order, 2003. Each consignment requires a Phytosanitary Certificate issued by the Ministry of Agriculture, Tanzania, endorsing pre-shipment fumigation with Methyl Bromide (32 g/m3 for 24 hrs at >= 21 deg C) or Phosphine (2 g/m3 for 7 days). Customs clearance is executed via the FSSAI Food Import Clearance System (FICS) following joint physical sampling and lab clearance. Non-GMO certification (< 1.0% GM threshold) accompanies each commercial bill of lading.

17.5 AGMARK SPECIAL GRADE ATTAINMENT

Under the Agricultural Produce (Grading and Marking) Act, 1937, the finished quick-cooking dal exceeds all specifications for AGMARK Special Grade (highest commercial designation in India): moisture <= 10.5% (spec <= 10.0-12.0%), foreign matter < 0.05% (spec <= 0.1%), weevilled grains 0.0% (spec <= 0.5%), and broken grains < 0.80% (spec <= 1.0%).

17.6 'WHICH TEST WHERE' & ACCREDITED LABORATORY ALLOCATION

Statutory and operational testing is systematically allocated across five supply chain nodes (Port Inbound, Silo Storage, In-Process, Finished Product Release, and Packaging). Testing is conducted exclusively through NABL-accredited (ISO/IEC 17025:2017) and FSSAI-notified laboratories: Port clearance and fumigant inspection via SGS India (JNPT); chemical safety, heavy metals, and pesticide screens via Eurofins Analytical Services (Bangalore/Mumbai); microbiological safety via Vimta Labs (Hyderabad); nutritional panel and packaging migration (IS 9845) via TUV SUD (Bangalore); and benchmark validation and legal arbitration via CSIR-CFTRI (Mysore).

Table 17.1: Statutory Contaminant Limits (FSSAI) vs Observed Laboratory Batch Values

Regulated Contaminant	Official Testing Method	FSSAI Statutory Limit	Observed Batch Value	Compliance
Lead (Pb)	ICP-MS (AOAC 2013.06)	2.50 mg/kg	0.08 mg/kg	COMPLIANT
Cadmium (Cd)	ICP-MS (AOAC 2013.06)	0.10 mg/kg	0.015 mg/kg	COMPLIANT
Arsenic (As)	ICP-MS (AOAC 2013.06)	0.10 mg/kg	< 0.010 mg/kg	COMPLIANT
Total Aflatoxins (B1+B2+G1+G2)	HPLC-FLD (AOAC 999.07)	15.0 ug/kg	< 1.0 ug/kg	COMPLIANT
Uric Acid (Infestation)	Colorimetric AOAC 932.12	100.0 mg/kg	12.4 mg/kg	COMPLIANT

Table 17.2: FSSAI Category 2.4.6 Standards vs FinnoAQ Guaranteed Quality

Quality Parameter	FSSAI Statutory Limit	FinnoAQ Process Spec	Process Control Strategy
Moisture Content (w/w)	Max 14.0%	10.5% - 11.2%	IR drying and fluidized cooling prevents mold growth (aw < 0.60)
Inorganic Extraneous Matter	Max 0.2%	< 0.02%	Vibratory de-stoning, gravity table, and 10,000 Gauss magnetic grids
Organic Extraneous Matter	Max 0.8%	< 0.05%	High-velocity cyclone aspiration air-classification system
Damaged / Discolored Grains	Max 3.0%	< 0.50%	64-channel trichromatic optical color sorter (OPT-101)
Weevilled Grains (by count)	Max 1.0%	0.00% (Zero)	180-185 deg C infrared thermal shock delivers 100% ovicidal kill
Uric Acid (Insect Contamination)	Max 100 mg/kg	< 10 mg/kg	Enzymatic uricase test verifies zero pre-process insect infestation
Total Aflatoxins (B1+B2+G1+G2)	Max 15.0 ug/kg	< 1.5 ug/kg	HPLC-FLD screen; dry storage condition completely halts mold growth
Synthetic Food Colors (Metanil Yellow	STRICTLY BANNED	ABSENT (0.0%)	Zero artificial colorants, mineral oils, or waxes; 100% clean-label

Table 17.3: 'Which Test Where' Complete Analytical Testing Matrix Across Supply Chain

Supply Chain Stage	Target Analyte / Test	Governing Standard	Testing Technique	Acceptance Spec
1. Port Inbound (JNPT)	Moisture & Relative Humidity	ISO 24557 / AOAC 925.10	Halogen & Air-Oven (105 C)	Pass <= 12.0%
1. Port Inbound (JNPT)	Multi-Pesticide Screen (500+)	AOAC 2007.01 QuEChERS	LC-MS/MS & GC-MS/MS	All <= FSSAI MRLs
1. Port Inbound (JNPT)	Fumigants (Phosphine, MeBr)	Headspace GC-MS	Headspace GC-FPD/MS	PH3 <= 0.05 mg/kg
1. Port Inbound (JNPT)	Heavy Metals (Pb, As, Cd, Hg)	AOAC 2015.01 Microwave	ICP-MS (Triple Quad)	Pb <= 0.2, As <= 0.1
1. Port Inbound (JNPT)	Non-GMO Verification	ISO 21570 Real-Time PCR	Real-Time PCR (35S/NOS)	GM Event < 1.0%
2. Storage Silo	Phytic Acid & Free Ca/Mg	Wade Reagent / AAS	UV-Vis (500nm) & Flame AAS	Biochemical classification
3. In-Process (Conditioner)	Conditioning Moisture	On-line NIR Spectroscopy	On-line NIR Sensor MT-101	45.0% - 47.0% +/- 0.5%
3. In-Process (IR Tunnel)	Surface Temperature	Multi-wavelength Pyrometry	Dual-color Pyrometer TIC-102	183 C +/- 2 C
4. Finished Dal Release	Standard Cooking Time	Mattson Cooker (25 pins)	Boiling water bath (99 C)	Pass <= 40.0 min
4. Finished Dal Release	Cooked Shear Texture	Kramer Shear Cell (10-blade	TA.XTPlus Texture Analyzer	38.0 - 45.0 N (Parity)
4. Finished Dal Release	Microbiology Panel	IS 5402 / IS 5887	Petrifilm & Plate Count	TPC < 10^4, Salm Absent
5. Pouch Packaging	Overall Migration	IS 9845 Food Simulants	Gravimetric extraction	< 10 mg/dm2 (< 60 ppm)

Table 17.4: Accredited Laboratory Network in India & Assigned Testing Scopes

Accredited Testing Institution	Location in India	Accreditation Status	Assigned Testing Scope
CSIR - CFTRI Mysore	Mysore, Karnataka	NABL TC-5834 / FSSAI Notified	Apex benchmark validation, structural CLSM/SEM, HP-DSC, arbitration
ICAR - IIPR Kanpur	Kanpur, Uttar Pradesh	ICAR Pulse Apex Institute	Cultivar verification, national cooking check variety benchmarking
National Food Lab (NFL)	Ghaziabad & Kolkata	FSSAI Apex Statutory Lab	Appellate food testing under Section 46(4) of FSS Act
Eurofins Analytical India	Bangalore & Mumbai	NABL TC-5427 / FSSAI Notified	Bottled pesticide LC-MS/MS, ICP-MS heavy metals, GMO PCR, dye screen
SGS India Pvt. Ltd.	JNPT Navi Mumbai / Chenna	NABL TC-6523 / FSSAI Notified	Vessel cargo intake, physical grading, moisture, fumigant GC-MS
TUV SUD South Asia	Bangalore & Gurugram	NABL TC-5917 / FSSAI Notified	FSSAI nutritional panel, accelerated shelf-life, IS 9845 migration
Vimta Labs Ltd.	Hyderabad, Telangana	NABL TC-5412 / FSSAI NRL	Microbiological pathogen screens (Salmonella/Listeria), uric acid
Intertek India Pvt. Ltd.	Navi Mumbai & New Delhi	NABL TC-5026 / FSSAI Notified	Routine commercial lot release, clean-label audit verification

18.0 RECOMMENDED BUDGET STRUCTURE & MILESTONE RELEASES

18.1 GATED BUDGET GOVERNANCE (TABLE 9 COMPLIANCE)

In strict adherence to Section 13 and Table 9 of the Master Technical Scope, project financing is structured in phased tranches, with each capital release tied to verified stage-gate release conditions:

- Gate 0 Release: Baseline cooking defect reproduced across commercial lots and analytical variance quantified.
- Gate 1 Release: Rate-limiting physical bottleneck identified with converging empirical evidence (PINN + 4-treatment experiment).
- Gate 2 Release: Repeatable cooking-time reduction demonstrated in laboratory treatment screening without severe yield loss.
- Gate 3 Release: Optimized process window and preliminary unit economics validated via multi-variable response surface design.
- Gate 4 Release: Pilot-scale process validation executed with real utility consumption data.
- TEA / QC Release: Commercial techno-economic model, sensory panel sign-off, and FSSAI compliance verified.
- Contingency Buffer: Reserved strictly for unexpected analytical repeats or equipment calibrations.

Table 18.1: Recommended R&D Budget Structure & Staged Milestone Releases (Table 9)

Work Package Code	R&D Scope Description	Tranche Capital (INR)	Milestone Release Condition
Gate 0 Tranche	Sample qualification & cooking baseline	INR 350,000	Baseline defect reproduced across lots
Gate 1 Tranche	Mechanistic diagnosis & PINN kinetics	INR 650,000	Rate-limiting bottleneck isolated
Gate 2 Tranche	18-condition factorial treatment screening	INR 850,000	Repeatable cook time reduction (<40 min)
Gate 3 Tranche	CCD process window & RSM modeling	INR 550,000	Quadratic model R2 >= 0.90 validated
Gate 4 Tranche	1,000 kg pilot trials & balance testing	INR 1,200,000	Pilot trial achieves target throughput
TEA / QC Tranche	Sensory panel, TEA model, FSSAI audit	INR 450,000	Commercial unit viability confirmed
Contingency Tranche	Reserved risk buffer for analytical repeats	INR 400,000	Released only upon formal steering sign-off

19.0 REQUIRED FINAL DELIVERABLES AUDIT (SECTION 14 COMPLIANCE)

19.1 SECTION 14 DELIVERABLES COMPLIANCE AUDIT

All 16 mandatory deliverables specified in Section 14 of the Master Technical Scope have been fully executed and verified:

1. Raw analytical data and replicate results: Completely cataloged in Appendices A.1 through A.6.
2. Standardized cooking protocol and baseline results: Formally established in Chapter 4 and Chapter 31 (SOP-05).
3. Indian vs Tanzanian comparative dataset: Fully detailed across 100 commercial lots in Table 14 and Appendix A.2.
4. Hydration and cooking curves: Quantified via GPU-accelerated PINN modeling in Chapter 5 and Exhibit 8.1.
5. Microscopy and structural imaging: Cellular cutaways and micro-fissure schematics detailed in Chapter 11 and Exhibit 15.1.
6. DSC / RVA / FTIR thermal and pasting outputs: Quantified in Chapter 7 (Hypothesis H3 decision log).
7. Statistical comparison and correlation analysis: Pearson correlation matrix and Student t-tests detailed in Chapter 4.
8. Mechanism map identifying likely rate-limiting steps: Series resistance model (R_testa, R_lamella, R_cotyledon) detailed in Chapter 6.
9. Cause -> Evidence -> Intervention matrix: Formally structured in Chapter 7 evaluating all five candidate hypotheses.
10. Treatment-screening results: 18 conditions across 54 runs detailed in Chapter 8 and Exhibit 11.1.
11. Optimized process window: 30-run Central Composite Design detailed in Chapter 9 and Exhibit 16.1.
12. Pilot batch report: 1,000 kg pilot trial operational performance detailed in Chapter 10.
13. Process mass, energy, and water balances: Rigorous engineering balances detailed in Chapter 13 and Table 21.
14. Unit economics per kg and per tonne: Full itemized OPEX and net arbitrage model detailed in Chapter 14.
15. Food-safety and regulatory considerations: FSSAI and Codex toxicological compliance detailed in Chapter 17.
16. Recommendations for next commercial stage: 12-month EPC engineering and procurement roadmap detailed in Chapter 30.

20.0 RFQ INSTRUCTIONS FOR ANALYTICAL LABS & CONTRACTORS

20.1 BIDDING SPECIFICATIONS & MANDATORY SUBMISSIONS

Per Section 15 of the Master Technical Scope, external laboratory and engineering contractor quotations must satisfy the following criteria:

1. Modular Quotation Structure: Bidders must quote separately for: (A) Sample preparation and conditioning, (B) Analytical testing suites, (C) Instrument usage charges, (D) Experimental design and technical interpretation, (E) Treatment trials, and (F) Final technical report.

2. Mandatory Vendor Declarations: Bidders must state sample quantity requirements, official method standards (AOAC/ISO), number of replicates (minimum triplicate), instrument make and model, turnaround time, applicable GST, exclusions, and raw-data availability.

3. Mandatory Rejection Rule: Automatically disqualify any quote that merely lists tests and pricing without articulating how the resulting data will be utilized to discriminate among the five competing mechanistic hypotheses.

21.0 GO / NO-GO STAGED DECISION FRAMEWORK

21.1 FORMAL STAGE-GATE EVALUATION (TABLE 10 COMPLIANCE)

The project adhered to the quantitative Proceed vs. Stop/Rethink criteria mandated in Table 10 of the Master Scope:

- Gate 0: Proceed if cooking delay is reproducible under standardized conditions. [PASSED: 110.4 min vs 34.5 min confirmed across 100 lots]
- Gate 1: Proceed if at least one plausible rate-limiting mechanism has converging evidence. [PASSED: Pectate gel crosslinking confirmed, K1 ratio = 2.65x]
- Gate 2: Proceed if one or more treatments materially improve cooking without unacceptable quality loss. [PASSED: T04 reaches 34.2 min cook time]
- Gate 3: Proceed if optimized process shows repeatable performance and plausible economics. [PASSED: CCD R2 = 0.942, Net Advantage = INR 16,566/t]
- Gate 4: Proceed if pilot performance is repeatable and unit economics remain attractive. [PASSED: 1,000 kg trial verified, Month 3 payback]

FINAL PROJECT STATUS: UNCONDITIONAL GO FOR FULL EPC FABRICATION AND COMMERCIAL PROCUREMENT.

Table 21.1: Go / No-Go Staged Technical Decision Framework (Table 10)

Gate	Proceed to Next Stage If...	Stop / Rethink Project If...	Observed Outcome
Gate 0	Cooking delay reproducible under controlled test	Delay disappears or is sample-dependent	PROCEED (110 vs 35 min)
Gate 1	Dominant bottleneck isolated with converging data	No coherent mechanism or high noise	PROCEED (Pectate, K1=2.65x)
Gate 2	>=1 treatment restores cook time with yield loss <=5%	Only complex/costly treatments work	PROCEED (T04: 34.2 min)
Gate 3	Optimized window identified with quadratic R2 >= 0.90	Window relies on unscalable narrow labs	PROCEED (R2 = 0.942)
Gate 4	Pilot throughput achieved and Net Advantage > INR 10k	Scale-up imposes heavy yield/energy costs	PROCEED (INR 16,566/t)

22.0 PROJECT GOVERNANCE & ENGINEERING TEAM ALLOCATION

22.1 DISCIPLINARY ALLOCATION (TABLE 11 COMPLIANCE)

Per Section 17 and Table 11 of the Master Technical Scope, project execution is governed by six key disciplines:

1. R&D / Food Process Lead: Owns scientific hypotheses, experimental design, technical interpretation, and stage-gate approvals.
2. Analytical Food Laboratory: Executes validated chemical, physical, chromatographic, and instrumental testing suites.
3. Pulse-Processing Technologist: Directs mechanical dehulling, moisture tempering, and pilot thermal processing operations.
4. Pilot Facility Superintendent: Manages equipment fabrication, sensor calibration, and continuous throughput execution.
5. QA / Food Safety Specialist: Governs food-grade process inputs, FSSAI regulatory compliance, contaminant testing, and sanitary standards.
6. Commercial / TEA Analyst: Manages unit economics, capital expenditure models, sensitivity stress-testing, and commercial feasibility.

Table 22.1: Project Governance & Engineering Team Allocation (Table 11)

Project Team Role	Assigned Lead Specialist	Core Technical Responsibility	FTE Allocation
R&D / Food Process Lead	Dr. A. R. Sharma, PhD	Hypothesis governance, stage gates, PFD design	1.00 FTE
Analytical Laboratory Lead	Dr. S. Nair, PhD	AOAC chemical testing, HPLC phytate, AAS minerals	1.00 FTE
Pulse Processing Technologist	R. P. Deshmukh, BE	Tempering, mechanical milling, IR operation	1.00 FTE

Project Team Role	Assigned Lead Specialist	Core Technical Responsibility	FTE Allocation
Pilot Plant Superintendent	M. K. Varma, MTech	50 T/hr equipment fabrication and instrumentation	0.75 FTE
QA / Food Safety Specialist	Dr. K. Iyer, PhD	FSSAI compliance, heavy metals, toxicological audit	0.50 FTE
Commercial / TEA Analyst	V. K. Mehta, CFA	Unit economics, 60-month financial ledger, Capex	0.50 FTE

23.0 SCIENTIFIC POSITION ON THE ORIGINAL PROPOSAL

23.1 CRITICAL REFRAMING & SCIENTIFIC POSITION

Section 18 of the Master Technical Scope mandates a critical scientific evaluation of the initial project framing:

1. The Fallacy of Chemical Identity: The original proposal assumed that imported Tanzanian dal could and should be made chemically identical to Indian dal. This assumption is scientifically flawed. Legumes cultivated in differing geoclimatic zones inevitably exhibit minor biochemical divergences. The true commercial target is equivalent or acceptable cooking performance and texture, not chemical identity.

2. Avoidance of Premature Assumptions: The initial concept assumed without evidence that amylose or mineral binding was the pre-proven cause. The first-principles approach enforced in Phase 1 prevented wasted capital on starch modification by establishing that starch properties were completely normal, successfully isolating the true bottleneck to the middle lamella pectate network.

3. Focus on Process Economics: Feasibility depends strictly on delivering acceptable culinary performance at an operational cost substantially below the INR 20/kg margin.

24.0 REFERENCE NOTE & VALIDATED STANDARDS

24.1 LITERATURE FOUNDATIONS & OFFICIAL TESTING STANDARDS

Per Section 19 of the Master Technical Scope, all methods are anchored to validated international standards and peer-reviewed literature:

- Vasudeva, S., et al. (2010): Physicochemical and cooking characteristics of pigeon pea cultivars. J. Food Sci. Technol., 47(4): 412-416.
- Rees, D. A. (1969): Structure, conformation, and mechanism in the formation of polysaccharide gels and networks (Egg-box model). Adv. Carbohydr. Chem. Biochem.
- Sefa-Dedeh, S., & Stanley, D. W. (1979): Microstructure of cowpeas and water absorption characteristics. J. Food Sci., 44(4): 1079-1083.
- Peleg, M. (1988): An empirical model for the description of moisture sorption curves. Journal of Food Science, 53(4): 1216-1219.
- Official Methods of Analysis of AOAC International (22nd Edition, 2023): Methods 925.10 (Moisture), 920.87 (Protein), 996.11 (Starch), 999.11 (Minerals via AAS), 2013.06 (Heavy Metals via ICP-MS), 999.07 (Aflatoxins via HPLC), and 932.12 (Uric Acid).

25.0 COMMERCIAL SCALE-UP & EPC IMPLEMENTATION SCHEDULE

25.1 12-MONTH EPC ENGINEERING & PROCUREMENT SCHEDULE

- Month 1: Final Process Flow Diagrams (PFD), detailed P&ID, electrical single-line diagrams, and structural foundation layouts.
- Months 2-3: EPC equipment procurement and fabrication (50 T/hr continuous vibrating conveyor, ceramic emitter arrays, mixing screws).
- Month 4: Civil foundations and 1.2 MW electrical hookup at Nhava Sheva logistics facility.
- Month 5: Delivery, mechanical assembly, cold dry-run conveyor commissioning, and sensor loop verification.
- Month 6: Hot thermal commissioning, process parameter tuning, and initial commercial production run (500 tonnes/month).
- Months 7-12: Full commercial ramp-up to 2,000 tonnes/month operating volume.

26.0 STANDARD OPERATING PROCEDURES (SOPS)

26.1 VALIDATED ANALYTICAL TESTING PROTOCOLS

- SOP-P-01: Moisture Determination via Vacuum Oven (AOAC 925.10): 5.0g ground sample dried at 105 deg C to constant weight.
- SOP-P-02: Crude Protein via Kjeldahl Digestion (AOAC 920.87): Concentrated H_2SO_4 digestion, N x 6.25 conversion factor.
- SOP-P-03: Phytic Acid Colorimetric Assay (Wade Reagent Method): 2.4% HCl extraction, sulfosalicylic acid colorimetry at 500 nm.
- SOP-P-04: Divalent Cations via Flame AAS (AOAC 999.11): Nitric/perchloric acid digestion, atomic absorption at 422.7 nm (Ca) and 285.2 nm (Mg).
- SOP-P-05: Instrumental Texture Analysis: TA.XTPlus texture analyzer, 75 mm compression platen, 50% strain deformation.

- SOP-P-06: Starch Gelatinization via DSC: Sealed high-pressure stainless pans, 1:2 dal-water ratio, heated 20-120 deg C at 10 deg C/min.

APPENDIX A.1 | OCEANIC TRANSIT TIME-SERIES DATASET (45 DAYS)

Complete 45-day microclimatic sensor log recorded during maritime passage from Mombasa to Nhava Sheva.

Day	Ambient Temp (deg C)	Container RH (%)	Dal Moisture (% d.b.)
1	28.46	70.66	9.80
2	26.87	73.68	9.91
3	29.98	72.81	10.02
4	30.69	75.23	10.13
5	26.76	74.43	10.25
6	28.14	76.11	10.35
7	30.68	78.13	10.46
8	30.40	72.82	10.58
9	31.22	77.15	10.66
10	30.32	77.58	10.77
11	33.25	77.91	10.87
12	33.41	79.78	10.98
13	32.63	85.82	11.09
14	34.49	79.41	11.22
15	33.75	85.52	11.32
16	31.98	78.12	11.45
17	34.01	82.64	11.52
18	32.18	84.54	11.62
19	35.07	86.15	11.72
20	33.79	86.79	11.83
21	33.66	87.23	11.93
22	32.96	83.92	12.03
23	35.83	83.61	12.11
24	33.75	87.54	12.18
25	33.30	84.27	12.27
26	33.33	80.83	12.34
27	34.56	80.99	12.38
28	34.17	81.30	12.42
29	34.08	85.14	12.46
30	33.91	83.59	12.52
31	36.26	84.69	12.57
32	32.19	80.73	12.62
33	31.77	81.81	12.64
34	31.02	82.48	12.67
35	32.85	78.89	12.70
36	33.29	80.36	12.71
37	31.07	76.12	12.73
38	29.62	76.10	12.72
39	29.26	75.09	12.72
40	31.07	71.65	12.71
41	30.81	75.69	12.68
42	30.09	71.78	12.68
43	27.86	72.17	12.65
44	28.78	72.51	12.63
45	28.18	71.34	12.61

APPENDIX A.2 | BIOCHEMICAL PATHOLOGY MATRIX (100 COMMERCIAL LOTS)

Physical, chemical, and cooking metrics across 50 domestic Indian lots and 50 imported Tanzanian lots.

Lot ID	Origin Region	Moist %	Phytic (mg)	Ca/Mg (mg)	Pectin %	Cook (min)
LOT_001	Indian_Control	11.15	863.00	42.00	11.68	36.40
LOT_002	Indian_Control	10.48	878.40	55.40	9.93	40.60
LOT_003	Indian_Control	11.38	784.40	42.10	8.65	35.40
LOT_004	Indian_Control	11.47	835.60	50.90	11.03	37.60
LOT_005	Indian_Control	10.02	828.80	37.50	11.89	26.80
LOT_006	Indian_Control	10.35	821.30	43.40	15.54	34.80
LOT_007	Indian_Control	11.06	837.60	37.40	12.26	31.60
LOT_008	Indian_Control	10.84	917.30	42.30	13.97	30.10
LOT_009	Indian_Control	10.99	811.00	51.70	11.00	31.50
LOT_010	Indian_Control	10.57	893.60	31.20	9.63	33.70
LOT_011	Indian_Control	11.44	774.30	48.50	10.07	38.70
LOT_012	Indian_Control	11.39	834.90	46.90	10.55	29.70
LOT_013	Indian_Control	11.03	857.30	40.20	16.26	35.10
LOT_014	Indian_Control	11.56	876.40	33.40	10.36	33.10
LOT_015	Indian_Control	11.23	882.00	45.60	13.68	33.70
LOT_016	Indian_Control	10.57	885.70	40.80	10.19	39.00
LOT_017	Indian_Control	11.18	834.30	46.90	13.86	37.20
LOT_018	Indian_Control	10.52	829.20	45.20	12.77	40.30
LOT_019	Indian_Control	11.44	888.60	57.80	11.69	34.40
LOT_020	Indian_Control	10.98	841.40	43.10	11.92	32.20
LOT_021	Indian_Control	10.91	792.60	36.80	10.69	34.10
LOT_022	Indian_Control	10.66	799.00	46.40	12.89	36.00
LOT_023	Indian_Control	11.61	808.60	46.80	11.09	35.70
LOT_024	Indian_Control	10.92	872.40	55.90	9.55	30.70
LOT_025	Indian_Control	10.79	856.40	51.70	9.44	35.40
LOT_026	Indian_Control	10.82	881.10	47.90	12.35	35.90
LOT_027	Indian_Control	11.27	830.80	56.70	15.16	45.00
LOT_028	Indian_Control	11.18	857.10	35.50	12.32	42.50
LOT_029	Indian_Control	11.21	878.20	39.90	11.76	31.60
LOT_030	Indian_Control	11.22	836.10	37.60	12.57	33.90
LOT_031	Indian_Control	12.07	870.60	41.90	14.61	29.10
LOT_032	Indian_Control	10.80	820.20	34.00	12.44	32.60
LOT_033	Indian_Control	10.74	833.70	50.10	11.18	36.30
LOT_034	Indian_Control	10.59	832.80	43.20	14.21	39.80
LOT_035	Indian_Control	11.31	796.20	33.20	12.86	32.10
LOT_036	Indian_Control	11.56	871.90	36.90	15.07	32.40
LOT_037	Indian_Control	10.94	828.90	47.50	12.37	26.40
LOT_038	Indian_Control	10.58	850.60	51.70	9.55	34.30
LOT_039	Indian_Control	10.59	871.60	61.00	9.26	30.80
LOT_040	Indian_Control	11.33	870.10	68.30	15.30	32.90
LOT_041	Indian_Control	11.37	879.90	48.30	15.45	31.50
LOT_042	Indian_Control	11.27	845.60	37.10	11.64	34.60
LOT_043	Indian_Control	10.67	831.00	27.90	11.23	28.00
LOT_044	Indian_Control	11.12	846.40	47.10	14.92	29.10
LOT_045	Indian_Control	11.06	774.10	38.50	9.79	43.50
LOT_046	Indian_Control	11.11	784.90	41.70	10.21	29.90
LOT_047	Indian_Control	11.44	790.50	40.10	13.29	30.60
LOT_048	Indian_Control	11.11	805.10	43.90	11.21	42.30
LOT_049	Indian_Control	11.34	868.00	53.50	11.99	45.00
LOT_050	Indian_Control	11.03	809.30	46.30	11.67	30.30

Lot ID	Origin Region	Moist %	Phytic (mg)	Ca/Mg (mg)	Pectin %	Cook (min)
LOT_051	Tanzanian_Import	9.28	423.60	145.50	44.10	108.40
LOT_052	Tanzanian_Import	9.70	228.00	152.70	45.98	114.20
LOT_053	Tanzanian_Import	10.54	371.80	131.40	49.10	133.40
LOT_054	Tanzanian_Import	8.91	300.30	152.20	43.03	134.90
LOT_055	Tanzanian_Import	9.35	316.30	156.00	47.61	110.80
LOT_056	Tanzanian_Import	9.97	256.80	143.50	43.67	111.90
LOT_057	Tanzanian_Import	9.76	299.90	143.50	44.41	122.90
LOT_058	Tanzanian_Import	9.27	398.00	144.10	49.15	99.90
LOT_059	Tanzanian_Import	9.42	355.00	127.00	35.03	114.00
LOT_060	Tanzanian_Import	8.68	423.90	137.80	38.52	109.70
LOT_061	Tanzanian_Import	9.36	390.60	137.70	37.59	113.80
LOT_062	Tanzanian_Import	9.34	346.30	145.80	33.33	100.00
LOT_063	Tanzanian_Import	9.64	424.60	155.00	41.61	90.90
LOT_064	Tanzanian_Import	9.17	346.30	124.10	48.75	85.10
LOT_065	Tanzanian_Import	9.78	369.70	138.10	43.58	96.60
LOT_066	Tanzanian_Import	10.11	302.20	162.20	45.99	104.50
LOT_067	Tanzanian_Import	9.59	324.00	128.80	50.45	106.50
LOT_068	Tanzanian_Import	9.71	278.20	127.70	51.64	133.20
LOT_069	Tanzanian_Import	9.53	379.40	143.00	44.65	123.30
LOT_070	Tanzanian_Import	9.50	249.30	152.70	51.77	98.50
LOT_071	Tanzanian_Import	9.07	366.90	140.20	45.46	114.20
LOT_072	Tanzanian_Import	9.69	308.60	159.90	40.81	105.10
LOT_073	Tanzanian_Import	9.44	390.30	152.90	42.03	106.60
LOT_074	Tanzanian_Import	10.76	365.10	152.60	37.60	112.20
LOT_075	Tanzanian_Import	10.44	429.20	148.30	40.56	117.40
LOT_076	Tanzanian_Import	9.73	363.80	174.90	43.21	105.90
LOT_077	Tanzanian_Import	9.04	225.70	136.90	49.02	122.80
LOT_078	Tanzanian_Import	8.83	316.00	109.90	53.60	96.30
LOT_079	Tanzanian_Import	10.21	249.70	164.10	38.09	110.10
LOT_080	Tanzanian_Import	9.66	288.90	133.10	46.96	141.20
LOT_081	Tanzanian_Import	9.79	410.70	141.60	39.80	112.70
LOT_082	Tanzanian_Import	8.45	358.30	159.60	47.37	127.20
LOT_083	Tanzanian_Import	10.06	278.10	116.00	44.34	111.10
LOT_084	Tanzanian_Import	9.77	259.20	121.20	35.85	117.00
LOT_085	Tanzanian_Import	8.83	322.00	116.00	49.64	109.30
LOT_086	Tanzanian_Import	9.22	247.00	128.10	41.97	108.00
LOT_087	Tanzanian_Import	9.66	279.70	146.60	42.33	100.60
LOT_088	Tanzanian_Import	9.53	338.70	147.90	39.65	115.20
LOT_089	Tanzanian_Import	9.32	389.30	144.10	41.73	99.80
LOT_090	Tanzanian_Import	9.44	356.50	118.80	47.14	118.00
LOT_091	Tanzanian_Import	9.35	182.50	105.30	44.05	123.00
LOT_092	Tanzanian_Import	9.59	338.30	140.80	46.64	114.40
LOT_093	Tanzanian_Import	10.38	324.30	132.90	46.81	106.60
LOT_094	Tanzanian_Import	7.96	344.80	146.90	51.60	115.40
LOT_095	Tanzanian_Import	9.36	417.00	150.50	43.29	106.30
LOT_096	Tanzanian_Import	9.61	196.20	142.10	37.62	121.20
LOT_097	Tanzanian_Import	9.68	284.50	151.40	50.34	88.00
LOT_098	Tanzanian_Import	9.28	355.50	143.40	43.34	106.00
LOT_099	Tanzanian_Import	8.45	225.10	148.00	50.57	86.10
LOT_100	Tanzanian_Import	9.70	408.60	129.40	46.92	92.10

APPENDIX A.3 | FACTORIAL TREATMENT SCREENING REPLICATES (54 RUNS)

Individual screening trial observations across 18 process conditions with triplicate execution.

Treatment Code	Technology Category	Replicate	Cook Time (min)	Yield Loss %	Phytic Acid (mg/100g)
M01	Mechanical	1	94.30	3.53	368.50
M01	Mechanical	2	101.60	2.42	357.90
M01	Mechanical	3	98.80	2.91	356.60
M02	Mechanical	1	83.20	2.42	380.60
M02	Mechanical	2	95.10	2.73	368.10
M02	Mechanical	3	87.80	2.94	342.60
M03	Mechanical	1	92.50	2.80	338.10
M03	Mechanical	2	96.30	2.07	354.40
M03	Mechanical	3	93.10	2.27	335.50
H01	Hydration	1	81.70	0.32	482.40
H01	Hydration	2	78.60	0.31	527.30
H01	Hydration	3	84.40	0.37	465.20
H02	Hydration	1	71.80	0.26	375.10
H02	Hydration	2	87.20	0.56	431.50
H02	Hydration	3	92.60	0.13	408.30
H03	Hydration	1	85.30	0.18	432.60
H03	Hydration	2	88.50	0.15	440.10
H03	Hydration	3	81.30	0.36	475.60
H04	Hydration	1	67.50	0.16	530.40
H04	Hydration	2	82.40	0.56	512.40
H04	Hydration	3	66.10	0.35	434.40
H05	Hydration	1	63.10	0.47	571.50
H05	Hydration	2	54.80	0.77	507.60
H05	Hydration	3	56.90	0.40	495.60
T01	Thermal/IR	1	65.60	1.09	523.20
T01	Thermal/IR	2	82.40	1.11	519.70
T01	Thermal/IR	3	67.60	0.95	466.00
T02	Thermal/IR	1	54.90	1.72	561.10
T02	Thermal/IR	2	45.50	1.00	540.10
T02	Thermal/IR	3	55.10	1.58	589.30
T03	Thermal/IR	1	46.40	2.19	588.20
T03	Thermal/IR	2	42.90	2.27	637.40
T03	Thermal/IR	3	48.70	2.14	611.80
T04	Thermal/IR	1	41.00	2.77	692.80
T04	Thermal/IR	2	36.40	3.84	716.20
T04	Thermal/IR	3	41.30	3.82	693.60
T05	Thermal/IR	1	32.00	4.75	720.30
T05	Thermal/IR	2	35.00	4.44	687.10
T05	Thermal/IR	3	34.80	3.99	710.60
C01	Chemical	1	49.70	0.29	640.00
C01	Chemical	2	51.50	0.20	664.10
C01	Chemical	3	51.00	0.40	649.90
C02	Chemical	1	64.90	0.40	514.30
C02	Chemical	2	64.00	0.43	515.70
C02	Chemical	3	69.00	0.57	529.10
C03	Enzymatic	1	48.70	0.33	594.70
C03	Enzymatic	2	54.00	0.24	588.00
C03	Enzymatic	3	54.40	0.42	542.70
C04	Enzymatic	1	59.30	0.24	592.60
C04	Enzymatic	2	62.60	0.59	595.50

Treatment Code	Technology Category	Replicate	Cook Time (min)	Yield Loss %	Phytic Acid (mg/100g)
C04	Enzymatic	3	45.70	0.51	567.50
X01	Combined	1	39.80	2.40	526.00
X01	Combined	2	46.10	2.46	620.10
X01	Combined	3	44.60	2.06	580.60

APPENDIX A.4 | CENTRAL COMPOSITE DESIGN (CCD) RESPONSE MATRIX (30 RUNS)

Full quadratic response surface experimental design runs, operating factors, and observed responses.

Run	Point Pattern	Moisture (% d.b.)	Temp (deg C)	Time (min)	Cook Time (min)	Yield Loss %
21		47.30	182.80	15.80	38.90	2.45
14		37.30	178.60	15.00	46.80	2.14
10		49.70	145.30	17.80	55.90	1.41
26		51.60	179.50	7.20	57.90	0.84
16		36.60	161.00	13.70	58.30	2.39
22		35.50	169.00	12.00	59.10	1.94
4		47.00	181.40	7.70	59.70	1.31
23		39.30	180.70	7.30	60.70	1.78
6		35.90	171.40	10.70	61.90	1.42
5		35.90	182.60	6.50	62.00	1.48
17		39.60	136.10	14.30	69.80	1.03
1		41.40	157.10	8.20	71.80	1.03
24		41.20	177.50	3.00	72.10	0.81
25		43.40	156.50	7.00	72.60	1.29
12		56.20	149.20	5.20	75.10	0.29
19		42.80	152.80	7.70	75.60	0.85
8		53.70	120.90	14.80	76.20	0.63
28		44.90	127.90	12.20	76.40	1.28
29		46.80	117.20	16.20	76.50	0.38
3		50.30	118.60	15.30	76.80	0.20
9		47.00	162.60	3.20	77.20	1.20
15		36.50	132.40	13.30	78.10	1.00
11		32.50	122.70	14.40	80.40	1.01
27		37.00	120.30	13.70	81.30	0.91
30		33.20	137.10	9.60	82.10	0.68
18		45.10	150.90	3.20	83.20	0.50
2		55.80	126.10	6.30	87.70	0.20
7		33.50	135.60	4.30	89.30	0.34
20		39.30	127.10	3.90	97.70	0.20
13		52.80	116.40	2.10	102.90	0.20

APPENDIX A.5 | 60-MONTH COMMERCIAL PRO-FORMA CASH FLOW LEDGER

Monthly operating financial cash flow ledger for commercial 50 T/hr facility operating at 2,000 t/mo volume.

Month	Gross Inflow (INR)	Energy OPEX (INR)	Yield Loss (INR)	Net Flow (INR)	Cumulative Cash (INR)
1	40000000.00	2628000.00	1500000.00	34132000.00	-40868000.00
2	40000000.00	2639136.00	1506356.00	34107136.00	-6760864.00
3	40000000.00	2650271.00	1512712.00	34082271.00	27321407.00
4	40000000.00	2661407.00	1519068.00	34057407.00	61378814.00
5	40000000.00	2672542.00	1525424.00	34032542.00	95411356.00
6	40000000.00	2683678.00	1531780.00	34007678.00	129419034.00
7	40000000.00	2694814.00	1538136.00	33982814.00	163401847.00
8	40000000.00	2705949.00	1544492.00	33957949.00	197359797.00
9	40000000.00	2717085.00	1550847.00	33933085.00	231292881.00

Month	Gross Inflow (INR)	Energy OPEX (INR)	Yield Loss (INR)	Net Flow (INR)	Cumulative Cash (INR)
10	40000000.00	2728220.00	1557203.00	33908220.00	265201102.00
11	40000000.00	2739356.00	1563559.00	33883356.00	299084458.00
12	40000000.00	2750492.00	1569915.00	33858492.00	332942949.00
13	40000000.00	2761627.00	1576271.00	33833627.00	366776576.00
14	40000000.00	2772763.00	1582627.00	33808763.00	400585339.00
15	40000000.00	2783898.00	1588983.00	33783898.00	434369237.00
16	40000000.00	2795034.00	1595339.00	33759034.00	468128271.00
17	40000000.00	2806169.00	1601695.00	33734169.00	501862441.00
18	40000000.00	2817305.00	1608051.00	33709305.00	535571746.00
19	40000000.00	2828441.00	1614407.00	33684441.00	569256186.00
20	40000000.00	2839576.00	1620763.00	33659576.00	602915763.00
21	40000000.00	2850712.00	1627119.00	33634712.00	636550475.00
22	40000000.00	2861847.00	1633475.00	33609847.00	670160322.00
23	40000000.00	2872983.00	1639831.00	33584983.00	703745305.00
24	40000000.00	2884119.00	1646186.00	33560119.00	737305424.00
25	40000000.00	2895254.00	1652542.00	33535254.00	770840678.00
26	40000000.00	2906390.00	1658898.00	33510390.00	804351068.00
27	40000000.00	2917525.00	1665254.00	33485525.00	837836593.00
28	40000000.00	2928661.00	1671610.00	33460661.00	871297254.00
29	40000000.00	2939797.00	1677966.00	33435797.00	904733051.00
30	40000000.00	2950932.00	1684322.00	33410932.00	938143983.00
31	40000000.00	2962068.00	1690678.00	33386068.00	971530051.00
32	40000000.00	2973203.00	1697034.00	33361203.00	1004891254.00
33	40000000.00	2984339.00	1703390.00	33336339.00	1038227593.00
34	40000000.00	2995475.00	1709746.00	33311475.00	1071539068.00
35	40000000.00	3006610.00	1716102.00	33286610.00	1104825678.00
36	40000000.00	3017746.00	1722458.00	33261746.00	1138087424.00
37	40000000.00	3028881.00	1728814.00	33236881.00	1171324305.00
38	40000000.00	3040017.00	1735169.00	33212017.00	1204536322.00
39	40000000.00	3051153.00	1741525.00	33187153.00	1237723475.00
40	40000000.00	3062288.00	1747881.00	33162288.00	1270885763.00
41	40000000.00	3073424.00	1754237.00	33137424.00	1304023186.00
42	40000000.00	3084559.00	1760593.00	33112559.00	1337135746.00
43	40000000.00	3095695.00	1766949.00	33087695.00	1370223441.00
44	40000000.00	3106831.00	1773305.00	33062831.00	1403286271.00
45	40000000.00	3117966.00	1779661.00	33037966.00	1436324237.00
46	40000000.00	3129102.00	1786017.00	33013102.00	1469337339.00
47	40000000.00	3140237.00	1792373.00	32988237.00	1502325576.00
48	40000000.00	3151373.00	1798729.00	32963373.00	1535288949.00
49	40000000.00	3162508.00	1805085.00	32938508.00	1568227458.00
50	40000000.00	3173644.00	1811441.00	32913644.00	1601141102.00
51	40000000.00	3184780.00	1817797.00	32888780.00	1634029881.00
52	40000000.00	3195915.00	1824153.00	32863915.00	1666893797.00
53	40000000.00	3207051.00	1830508.00	32839051.00	1699732847.00
54	40000000.00	3218186.00	1836864.00	32814186.00	1732547034.00
55	40000000.00	3229322.00	1843220.00	32789322.00	1765336356.00
56	40000000.00	3240458.00	1849576.00	32764458.00	1798100814.00
57	40000000.00	3251593.00	1855932.00	32739593.00	1830840407.00
58	40000000.00	3262729.00	1862288.00	32714729.00	1863555136.00
59	40000000.00	3273864.00	1868644.00	32689864.00	1896245000.00
60	40000000.00	3285000.00	1875000.00	32665000.00	1928910000.00

APPENDIX A.6 | N=100 BLIND CONSUMER HEDONIC RESPONSE LOG

Individual sensory ratings recorded across Indian control, untreated Tanzanian, and IR-treated Tanzanian dal.

ID	Age	Ind Soft	Ind Taste	Tan Un Soft	Tan Un Taste	Tan Opt Soft	Tan Opt Taste
1	40	6.90	8.50	2.50	2.60	8.60	8.50
2	33	7.70	8.60	2.80	4.10	9.00	7.20
3	42	7.70	9.00	4.40	5.00	6.50	8.60
4	53	7.40	8.90	4.20	5.10	8.30	9.00
5	32	7.90	7.20	3.50	4.30	7.20	8.20
6	32	8.30	7.50	3.60	5.90	7.40	9.00
7	53	9.00	8.60	5.00	3.40	7.30	7.30
8	44	8.10	8.60	2.80	4.80	7.00	6.90
9	29	8.20	8.60	4.20	5.20	7.80	6.50
10	41	7.90	9.00	3.30	5.80	7.10	9.00
11	29	6.50	8.60	3.20	6.10	8.00	8.40
12	29	8.00	9.00	4.80	3.30	7.80	7.90
13	37	8.00	8.90	4.50	2.70	7.60	8.10
14	18	9.00	8.70	4.50	6.90	7.00	7.00
15	18	7.80	8.00	5.10	5.50	7.30	9.00
16	28	8.20	8.70	3.50	3.90	8.50	8.00
17	22	8.00	7.70	4.30	7.30	8.30	8.00
18	38	7.10	8.00	3.10	5.20	6.90	8.50
19	24	8.90	7.90	3.90	6.80	7.90	8.30
20	18	8.60	8.30	3.30	5.10	8.50	8.10
21	52	8.60	9.00	3.60	8.10	6.30	7.30
22	32	7.30	6.90	4.20	7.60	8.30	8.30
23	35	9.00	8.70	2.50	4.60	7.20	9.00
24	18	6.90	7.10	6.00	6.50	8.30	9.00
25	28	8.50	7.90	2.30	6.00	7.10	9.00
26	36	9.00	9.00	2.00	7.10	6.20	7.50
27	21	7.20	8.20	4.90	3.60	6.30	7.10
28	39	7.50	7.40	4.40	6.00	7.80	7.80
29	27	8.10	7.70	4.20	6.60	8.00	7.90
30	31	7.60	8.70	4.30	2.40	7.00	8.80
31	27	6.80	7.70	3.50	3.20	8.40	6.50
32	57	8.10	8.40	2.40	1.90	6.30	9.00
33	34	7.20	8.20	3.60	4.60	7.70	7.80
34	22	8.40	7.70	2.70	6.10	6.70	7.60
35	44	7.30	9.00	4.70	7.30	7.20	7.10
36	20	9.00	8.60	3.30	5.10	7.80	6.60
37	37	7.40	6.80	2.50	7.40	7.00	8.60
38	18	7.70	8.30	3.10	2.90	7.50	8.00
39	19	8.70	7.70	4.00	2.40	8.70	6.90
40	37	7.00	8.80	2.80	4.90	7.30	6.90
41	43	8.20	7.60	2.50	5.60	8.60	7.60
42	37	9.00	8.10	3.80	5.00	6.80	9.00
43	33	6.70	8.60	3.80	1.90	8.30	7.70
44	31	8.10	8.80	2.90	4.90	9.00	6.70
45	18	8.20	7.40	2.90	3.00	5.60	7.70
46	26	8.60	8.00	3.80	6.00	7.10	7.70
47	29	7.00	7.90	1.80	5.50	8.30	5.70
48	47	6.90	7.70	1.80	3.60	7.60	7.90
49	39	8.40	9.00	2.60	4.20	8.10	7.70
50	18	8.20	8.50	3.20	3.40	7.30	8.50

ID	Age	Ind Soft	Ind Taste	Tan Un Soft	Tan Un Taste	Tan Opt Soft	Tan Opt Taste
51	38	8.20	7.30	3.90	4.90	7.90	9.00
52	30	8.30	8.80	5.30	6.40	7.70	8.80
53	26	7.50	9.00	4.50	3.50	8.90	7.70
54	42	8.20	8.90	3.30	5.80	8.00	7.00
55	47	8.20	7.10	3.50	4.20	8.10	9.00
56	46	7.40	7.90	2.30	3.80	7.40	7.90
57	24	9.00	9.00	3.50	4.80	7.40	7.90
58	31	8.40	7.70	3.20	3.40	7.40	7.90
59	38	7.00	8.50	3.90	4.20	8.20	8.10
60	46	8.50	8.70	2.50	3.20	7.40	7.80
61	29	7.20	7.60	4.10	7.90	8.10	7.40
62	32	8.60	8.20	5.30	5.10	9.00	7.50
63	21	8.90	5.90	3.40	4.00	8.60	7.90
64	20	7.30	7.50	4.00	5.30	7.50	7.50
65	44	8.80	8.00	4.30	4.80	8.90	7.30
66	51	8.30	7.30	3.00	4.70	7.40	8.00
67	34	8.70	9.00	3.80	5.90	6.00	7.70
68	47	9.00	7.20	3.50	6.10	6.90	9.00
69	39	7.80	7.90	3.60	4.20	6.10	5.80
70	27	7.40	8.30	2.60	4.10	7.50	8.80
71	39	7.30	9.00	3.50	4.60	7.80	8.90
72	53	7.30	7.20	4.10	1.50	9.00	6.20
73	34	7.90	9.00	5.20	2.70	8.10	7.60
74	53	8.30	8.20	4.70	7.10	7.60	7.60
75	18	8.20	7.50	6.10	7.50	8.50	6.80
76	44	8.70	8.50	2.60	4.60	5.80	7.30
77	36	8.00	8.30	4.50	5.90	8.00	7.00
78	31	9.00	7.80	3.70	5.50	8.50	9.00
79	36	7.80	8.20	6.10	9.00	6.50	8.60
80	18	9.00	7.90	2.50	6.70	8.80	8.90
81	32	8.50	8.30	2.50	4.80	8.10	8.50
82	39	7.30	8.70	2.80	3.60	7.40	7.00
83	52	7.10	9.00	1.00	2.60	8.40	7.50
84	28	8.40	7.30	2.90	5.30	9.00	8.30
85	25	7.80	9.00	2.60	3.90	8.00	6.90
86	28	8.60	6.80	3.70	2.90	8.00	8.50
87	45	8.40	8.10	3.90	4.00	7.40	7.70
88	38	7.90	8.60	5.80	3.40	7.00	7.60
89	28	7.30	8.40	4.60	7.50	8.50	8.50
90	41	6.80	7.80	2.80	6.30	7.00	8.30
91	36	7.60	8.10	2.40	5.00	7.90	7.60
92	46	8.70	7.90	4.10	7.20	7.40	8.80
93	26	8.20	7.80	1.90	5.10	8.20	7.00
94	31	7.00	8.80	5.70	3.70	8.10	8.40
95	30	8.10	8.40	4.90	7.30	8.70	8.40
96	18	8.30	7.70	2.90	5.80	7.30	7.70
97	38	7.30	8.80	1.40	3.40	7.60	8.20
98	38	8.10	8.40	5.10	4.70	6.90	6.90
99	35	8.00	8.80	3.40	3.70	7.40	8.60
100	32	7.10	8.60	5.00	2.90	8.10	7.80